GRADUATION PROJECT | SYSTEM ANALYSIS AND DESIGN

SmartListing Copilot

Multimodal Listing Quality Control for Fashion E-Commerce

System Analysis and Design Document

UML structural and behavioral models, data and process analysis diagrams, use case descriptions, decision logic and schedule models, with full traceability to the requirements baseline.

Project	SmartListing Copilot, fashion listing quality control
Phase	Analysis and Design (milestone M1, 6 November 2026)
Baselines	Revised Scope proposal [1]; Planning [2]; Project Management [3]; Requirements [4]
Dataset	Fashion Product Images (Small), sole MVP dataset
Status	Final draft for team review; stakeholder approval pending
Date	29 October 2026

Scope: Product Type and Color only; five EDA-selected classes; controlled templates; seller Accept / Edit / Ignore; no external publishing.

Document Control

Item	Detail
Document	System Analysis and Design
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Prepared for	Graduation project team and academic review
Basis	Documents [1] to [4]; the Literature Review file was not available
Approval	Pending; no reviewer or supervisor sign-off is recorded
Change policy	Changes follow the change-control process of the Requirements document (section 18)

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1. Introduction and Document Basis

1.1 Purpose

This document is the System Analysis and Design deliverable of the SmartListing Copilot graduation project. It turns the planning, literature, requirements and project-management baselines into a complete set of analysis and design models: a problem statement, needs, stakeholders, feasibility, objectives, a system architecture blueprint, 27 required diagrams and the supporting explanations, tables and traceability that make each diagram usable by the team.

SmartListing Copilot is a proposed, pre-publication decision-support system for fashion e-commerce listings. It checks whether a product image agrees with the Product Type and Color claimed in the listing title and description, explains any contradiction with reason codes and severity, routes the listing to a three-state Review Queue, and proposes a controlled template correction that the seller alone may Accept, Edit or Ignore. Nothing is published to any external marketplace.

1.2 Scope of this document

- **In scope:** problem framing, needs, stakeholders, feasibility, objectives, architecture, all UML structural and behavioral diagrams, data-oriented and process-oriented analysis diagrams, use case descriptions with basic and alternative flows, decision logic tables, module and data models, scheduling diagrams (Gantt and PERT), and traceability to FR-001 to FR-025, NFR, UC-01 to UC-13 and OBJ-01 to OBJ-09.
- **Out of scope:** any new functional feature, any implementation, trained model, measured result or stakeholder approval. The designs describe the intended behavior of a proposed proof of concept.

1.3 Source documents and authority

Ref.	Document	Used for	Authority
[1]	SmartListing Copilot Project Proposal (Revised Scope), 12 Oct 2026	Scope, routing logic, constraints, cut order	Primary scope authority
[2]	Member 1 Project Planning Document (Revised), 16 Oct 2026	Problem statement, needs, objectives OBJ-01 to OBJ-09, stakeholders, feasibility	Objective baseline
[3]	Member 2 Project Management Document, version 1.0, 15 Oct 2026	WBS, Gantt, milestones, critical path, roles	Schedule baseline
[4]	Requirements Documentation (Parts I and II), 16 Oct 2026	FR-001 to FR-025, AC, NFR, UC-01 to UC-13, TAC, open decisions	Requirements baseline
[5]	Fashion Product Images (Small), Kaggle, P. Aggarwal	Sole MVP dataset (images and styles metadata)	Dataset definition

A Literature Review file was not part of the uploaded material (the second upload was byte-identical to the Planning and Management file), so this document draws on the other sources only. No literature claim is made here beyond what those documents already cite.

1.4 Evidence discipline

The source documents state that no model has been trained, no stakeholder interview or walkthrough has been held, no threshold has been frozen and no approval has been given. This document follows the same discipline:

- Numeric values in diagrams (for example the confidence 0.93 in the object diagram) are **illustrative examples**, never measured results.
- Open decisions (Q-01, Q-02, IN-01 to IN-04, D-01 to D-05) are shown as **To Be Confirmed** in the diagrams and tables, not resolved by assumption.
- Timing diagrams show ordering only; response-time targets are deferred (NFR-PERF-03).
- Findings of document-level consistency review are not stakeholder validation.

1.5 Conventions used in the diagrams

Convention	Meaning
Blue boxes / ellipses	Standard system elements: modules, use cases, processes
Green	Core or primary elements (Must Have use cases, CV or system-owned documents)
Purple	NLP-related elements
Orange	Extensions, optional features, flags and To Be Confirmed states
Red	Findings, mismatches and critical outcomes
Yellow	External entities, actors, notes and seller or reviewer actions
Grey	Stored data, configuration and local resources
IDs	FR-nnn functional requirement; NFR-xxx-nn non-functional; UC-nn use case; AC-nnn and TAC-nn acceptance criteria; OBJ-nn objective; Q-nn, IN-nn, D-nn open decisions; WBS numbers such as 3.3 follow the Project Management document

2. Problem Statement

2.1 Current situation

Fashion product information is often assembled and reviewed manually. The image, title and description of one listing can carry competing or incomplete representations of the same item, so a reviewer compares the three elements one by one to decide whether the listing is internally consistent. The Revised Scope describes this operational problem but provides no verified local time-study or marketplace incident data [1].

2.2 Identified problem

A listing may show one product type while naming another, describe a color that differs from the image, or omit a relevant type or color claim. These defects make listing quality difficult to judge consistently and create avoidable ambiguity for sellers and buyers. Manual detection is repetitive, and a generic warning without evidence gives little guidance on what to correct.

2.3 Desired situation

The desired state is an explainable pre-publication check that flags possible image-text disagreements, separates critical issues from uncertain cases, and proposes bounded text corrections while the seller keeps final control. Detecting inconsistency shifts attention to questionable listings and makes the basis of a suggested edit visible. It does not establish that returns, cost or review time will fall; such effects need separate real-world validation.

2.4 Problem-to-solution summary

Problem	Design response	Where modelled
Image names one type, text another	Parallel CV and NLP branches; comparison engine; TYPE_MISMATCH with severity	Sections 8.1, 9.4, 10.2, 10.9
Color in text differs from image	Color-family and near-shade rules; COLOR_MISMATCH	Sections 8.1, 10.7, 10.8
Reviewers cannot see where to look first	Three-state Review Queue and Inconsistency Level	Sections 9.3, 10.7, 10.8
Warnings without evidence	Reason codes, claimed vs predicted value, confidence, explanation	Sections 8.1, 8.2, 10.3
Risk of wrong or forced predictions	Low-confidence and unsupported-category handling; CLIP fallback; never forced labels	Sections 9.4, 9.2
Sellers need control over wording	Template suggestion with Before / After and Accept / Edit / Ignore; no publishing	Sections 9.1 (UC-10 to UC-12), 9.4

3. Needs Assessment

Needs are inferred from the proposal [1] and planning document [2]. They were not collected through interviews, surveys or field observation, and no legal or industry compliance requirement is asserted.

Business problem	Stakeholder need	Proposed system response	Expected benefit
Image-text inconsistency	Identify contradictions before a listing is published	Compare CV attribute predictions with claims extracted from title and description	Earlier visibility of questionable listing content
Incorrect product type	Know whether the named type fits the image	Predict articleType-level type within five classes; flag TYPE_MISMATCH	More consistent product-type review
Incorrect color information	Separate real color conflicts from near-shade differences	Predict baseColour-aligned color; normalise color families; COLOR_MISMATCH	More informative handling of color disagreements
Manual listing review	Direct attention to listings needing judgment	Review Queue: Ready to publish, Needs review, Critical mismatch	Prioritised checking instead of undifferentiated review
Unexplained warnings	See evidence and reasons behind a flag	Show predicted vs claimed attributes, confidence, severity and reason codes	Reviewable, interpretable findings
Seller-controlled corrections	Retain ownership of content changes	Template Before / After suggestion with Accept / Edit / Ignore	Actionable feedback without autonomous changes

3.1 Classification of needs

Class	Needs	Status
Felt needs (inferred)	Seller clarity, reviewer prioritisation, understandable warnings, control over corrections	Reasonable preferences, not confirmed by stakeholders
Normative / engineering-quality needs	Traceability, bounded outputs, explicit uncertainty, reproducible evaluation	Proposed technical quality expectations, not regulatory obligations
Needs to validate later	Terminology, warning comprehension, queue usefulness, editing behavior	Planned scenario walkthroughs (WT-01 to WT-06); none conducted

4. Stakeholder Analysis

Influence and impact are preliminary planning judgments, not empirically validated findings [2].

Stakeholder	Role	Interest	Expected needs	Influence	Impact
Marketplace sellers	Primary decision makers	Clear listings before publication	Simple submission, explainable flags, controlled edits	High	High
Catalog reviewers	Primary review users	Consistent content triage	Prioritised queue, severity, evidence	Medium	High
Development team	Build and evaluation	Implementable, bounded scope	Defined dataset, stable contracts, test cases	High	High
Operations / customer support	Indirect beneficiaries	Fewer ambiguous listings	Consistent listing-quality information	Low	Medium
Academic supervisor (if applicable)	Governance and assessment	Scope clarity, defensible evidence	Well-defined deliverables, academic integrity	High*	Medium*

* Conditional: the supervisor role requires confirmation in the academic setting.

4.1 Stakeholders in the system models

Stakeholder	System actor?	Use cases	Diagrams where it appears
Marketplace Seller	Yes, primary	UC-01, UC-09, UC-10, UC-12	Use case, activity, sequence, state, context, DFD, document flowchart
Catalog Reviewer	Yes, primary (review user)	UC-09, UC-13	Use case, sequence (queue), context, DFD, document flowchart
Development team	No (not a system role)	None	Package, deployment, component, Gantt, PERT
Operations / support	No (indirect beneficiary)	None	Not modelled as an actor
Academic supervisor	No (reviews documents only)	None	Open-decision register (section 11.3)

No administrator actor, account, authorization layer or admin dashboard exists in the approved MVP (NFR-SEC-09).

5. Feasibility Study

This is a pre-implementation assessment. It separates technology availability from dataset-specific and project-specific feasibility, which still need experiments [2].

5.1 Technical feasibility

- **Data:** Fashion Product Images (Small) provides images and a styles.csv map by product ID. articleType supervises Product Type, baseColour supervises Color and productDisplayName proxies the title. Class coverage, label ambiguity and the final five classes remain unverified until EDA (WBS 2.1).
- **Models:** ResNet and MobileNet transfer learning and CLIP zero-shot are established families. Suitability on this dataset is not yet demonstrated. Low resolution, similar shoe types, color ambiguity and unseen categories may limit reliability.
- **NLP and rules:** dictionary, regex and fuzzy matching with a bounded Arabic lexicon are feasible in standard Python. Whether the lightweight classifier improves on the rule baseline is an empirical question because mismatch cases are synthetic.
- **Integration:** FastAPI, Streamlit and Pydantic form a viable stack. A single data contract and early stubs reduce integration risk. CPU or GPU capacity, training time and weight-download access are not verified (A-04).

5.2 Operational feasibility

The seller provides an image and text and receives type and color evidence, uncertainty indications, a queue priority and a bounded suggestion. This can fit a pre-publication process because the seller keeps the final decision and reviewers focus on flagged cases. Fit is a design hypothesis: warning overload, confusing severity labels, unclear low-confidence messages, unrealistic synthetic text and narrow Arabic coverage could reduce usability. No user acceptance testing has been done.

5.3 Economic and resource feasibility

The stack is largely open source and the dataset is public, which can lower software-license barriers. This does not establish zero development or compute cost. No budget, return on investment or price is asserted. The six-person team has 588 forward planned hours across 44 work packages [3], and the schedule is tight relative to a multimodal build, so feasibility depends on protecting the core pipeline and keeping the cut order: Quality Score, Missing Attribute Detection, MLflow, then Docker.

5.4 Feasibility summary

Dimension	Evidence	Main constraints	Assessment	Validation needed
Technical	Dataset fields; proven model families; Python stack	Low resolution; category overlap; label noise; unknown runtime	Plausible, conditionally feasible	EDA; baselines; compute checks; integration tests
Operational	Human-controlled review; bounded warnings	No real seller study; false alarms unknown	Conceptually coherent	Scenario walkthroughs; queue testing
Economic / resources	Public dataset; open-source stack	Time, compute, storage and annotation effort undetermined	Potentially practical for a scoped academic prototype	Actual resource inventory; measured workloads
Schedule	Four fixed milestones; critical path defined	Tight window; training is the longest critical package	Feasible only with core protection	Weekly gate reviews (DG-1 to DG-3, IM-1 to IM-3)

6. Objectives and Scope

Overall objective: design, implement and evaluate a limited-scope, human-controlled multimodal workflow that identifies and explains Product Type and Color inconsistencies between fashion product images and listing text using Fashion Product Images Small [2].

ID	Objective	Primary design artifacts in this document
OBJ-01	Establish the end-to-end image + text listing-checking pipeline	Component, deployment, activity, sequence, interaction overview, flowchart, system flowchart
OBJ-02	Predict visual Product Type and Color for the selected five classes	Class (CV), structure chart (module 2), HIPO 2.0, DFD process 2.0
OBJ-03	Extract Product Type and Color claims from title / description	Class (NLP), structure chart (module 3), HIPO 3.0, DFD process 3.0
OBJ-04	Detect discrepancies between CV outputs and NLP claims	Class (comparison), DFD level 2 (4.0), decision tables, flowchart A
OBJ-05	Attach understandable evidence and reason codes to every warning	Object, ERD (FINDING, REASON_CODE), decision table (severity)
OBJ-06	Prioritise listings for human review	State machine, decision tree, decision tables, DFD level 2 (5.0), flowchart B
OBJ-07	Generate bounded suggested title / description corrections	Use cases UC-10 and UC-11, sequence (decision), HIPO 6.0
OBJ-08	Keep suggested edits under seller control	Use case UC-12, sequence (decision), state machine, document flowchart
OBJ-09	Evaluate the pipeline with transparent evidence and limitations	Timing diagram, Gantt, PERT, package (tests), traceability (section 11)

6.1 Scope boundaries

In scope (approved MVP)	Out of scope
One listing per request: image, title, description, category ResNet or MobileNet transfer learning plus CLIP zero-shot baseline and fallback English rules plus a small fixed Arabic lexicon Rule comparison plus a lightweight mismatch classifier Severity, reason codes, Inconsistency Level, three-state Review Queue Template suggestion, Before / After, Accept / Edit / Ignore FastAPI backend and Streamlit frontend; local demo only Supporting: Listing Quality Score, Missing Attribute Detection (cut-able)	Return prediction, dynamic pricing, price validation Material or size recognition LLM-generated content Autonomous publishing, live marketplace integration Production cloud deployment, batch CSV Broad multilingual NLP, additional commerce domains Competitor scraping, recommendation models LoRA fine-tuning, full multimodal transformers

6.2 Constraints that shape the designs

ID	Constraint	Effect on design
CON-01	Fashion Product Images Small is the sole dataset	Data store and deployment diagrams show only this dataset
CON-02	Exactly five articleType classes, chosen after EDA (not yet fixed)	Diagrams use generic type labels; the example class "Tshirts" is illustrative
CON-03	Only Product Type and Color are checked	Class, ERD and flowcharts contain no size, material or price attribute

ID	Constraint	Effect on design
CON-06	Templates only, no LLM	Template generator is a rule-based component
CON-07	Seller approval mandatory, no publishing	No publish interface or transition appears in any model
CON-09	One listing per request	No batch processing in activity, sequence or DFD
CON-11	Milestones 16 Oct, 6 Nov, 30 Nov, 4 Dec 2026	Gantt and PERT follow these dates

7. System Architecture Blueprint

7.1 Architectural overview

The system is a modular, local, three-tier prototype. A Streamlit presentation tier calls a FastAPI application tier over HTTP and JSON. The application tier validates input with a shared Pydantic data contract, runs the computer vision and NLP branches in parallel, then calls the comparison, review and template components in sequence and keeps results in an in-memory session store. Rules, vocabularies, thresholds and templates live in configuration files, and model weights are loaded once at start-up.

7.2 Tiers and responsibilities

Tier	Elements	Responsibility
Presentation	Streamlit UI	Intake form, results and findings, queue view, Before / After, Accept / Edit / Ignore. English UI with the Arabic limitation stated.
Application	FastAPI orchestrator, Pydantic contract, session store	Validation, parallel dispatch of branches, joining results, sequencing of downstream components, recording seller decisions, safe error handling.
Analysis components	CV module, NLP module, comparison engine, review engine, template generator, scoring module (optional)	Domain logic: visual prediction, claim extraction, mismatch detection, severity, queue routing, controlled suggestions.
Resources	Config store, model weights, local dataset	Frozen rules and thresholds, saved models, ID split lists. Read at start-up; no external service.

7.3 Module catalogue

Module	Interface	Main responsibilities	FRs
AnalysisOrchestrator	IAnalyzeApi	Validate listing, run branches in parallel, call engines, save and return AnalysisResult, record decisions	FR-001 to FR-003, FR-023 to FR-025
CVModule	ICvAnalysis	Image quality, Product Type and Color prediction with confidence, dominant-color cross-check, scope signals, CLIP baseline and fallback	FR-004 to FR-006, FR-013
NLPModule	INlpExtraction	Script detection, normalisation, Product Type and Color claim extraction (English rules, fixed Arabic lexicon)	FR-007, FR-008
ComparisonEngine	IComparison	Compare type and color, apply rules and lightweight classifier, severity, reason codes	FR-009 to FR-011, FR-016, FR-017
ReviewEngine	IReview	Low-confidence, blur, unsupported and missing-attribute checks; Inconsistency Level; queue status	FR-012, FR-014, FR-015, FR-019, FR-020
ScoringModule (optional)	IScoring	Transparent 0 to 100 score from image quality, completeness, consistency	FR-018
TemplateGenerator	ITemplateGeneration	Fill controlled templates with supported type and color only; Before / After change data	FR-021, FR-022
ConfigStore	files	Vocabularies, lexicons, rule tables, thresholds, templates (configuration, not code)	NFR-MNT-04, NFR-MNT-05

7.4 Key design decisions

Decision	Rationale	Source
CV and NLP branches run in parallel	Latency is bounded by the slower branch plus comparison, not the sum	NFR-PERF-02

Decision	Rationale	Source
Rules remain authoritative over the lightweight classifier	Deterministic, explainable behavior; the classifier supports but does not override	NFR-TEC-10, NFR-REL-03
CLIP is an independent baseline and fallback, not a preprocessing step	Allows disable-primary-model testing and scope disagreement detection	FR-004, NFR-REL-02
Uncertain, blurred and out-of-scope listings route to Needs review, never Ready	Honest handling of uncertainty is the product core	NFR-REL-01
No state or interface means "published"	Seller approval is mandatory and nothing leaves the machine	NFR-SEC-01, CON-07
One Pydantic data contract for every module	Stubs can be replaced without changing callers	NFR-MNT-02, NFR-MNT-03
A failed branch is isolated	The application keeps running and the listing is reported as Needs review	NFR-REL-04
Optional features removable by flag	Cut order Quality Score, Missing Attribute, MLflow, Docker keeps the core safe	NFR-MNT-06

7.5 Technology stack

Layer	Technology
Language	Python for ML, NLP and backend
Backend	FastAPI with Uvicorn; Pydantic data contract
Frontend	Streamlit
CV	ResNet or MobileNet (pretrained, transfer learning); CLIP zero-shot; resolution, blur and brightness checks; dominant color
NLP	Controlled vocabulary, regex, fuzzy matching, Arabic-character script detection
Comparison	Rule table (type groups, color families, near shades) plus logistic regression or small MLP
Evaluation	Accuracy, macro-F1, confusion matrix, ECE, precision, recall, F1, false-alarm rate, catch rate, response time
Optional	MLflow logging, Docker packaging, CLIP linear probe (only if the core is secure)

7.6 Architectural views and where to find them

View	Diagram	Section
Logical structure	Class diagrams, package diagram, component diagram	8.1, 8.6, 8.3
Data	Object diagram, ERD, DFD, CRUD matrix	8.2, 10.3, 10.2, 10.11
Deployment	Deployment diagram	8.4
Internal structure	Composite structure diagram, structure chart, HIPO	8.5, 10.9, 10.10
Behavior	Use case, activity, state, sequence, communication, interaction overview, timing	9.1 to 9.7
Decision logic	Flowcharts, decision tree, decision tables	10.4, 10.7, 10.8
Project view	Gantt chart, PERT network	10.12, 10.13

8. UML Structural Diagrams

Structural diagrams describe what the system is made of: classes and their relationships, runtime snapshots, components and their interfaces, the physical deployment, internal structure, code organisation and the stereotypes that tailor UML to this project. They follow UML 2.5.1 notation [6].

8.1 Class Diagram

The class diagrams define the static design of the proposed system. The first (shown in two parts) is the design-class view of modules, helpers and interfaces; the second is the data model of the Pydantic contract with its enumerations.

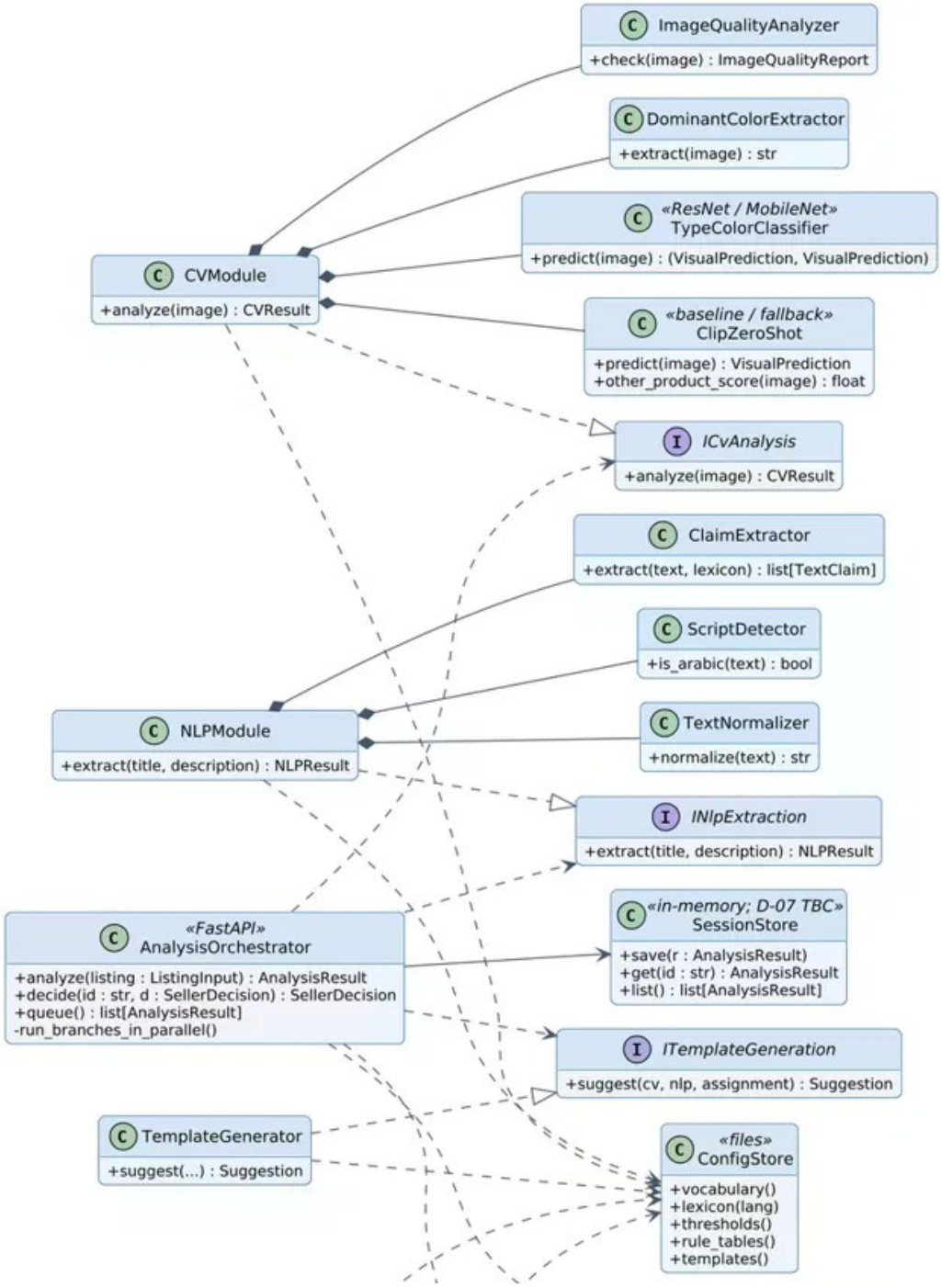


Figure 1. *Class diagram, design view (part 1 of 2): CV module, NLP module and orchestrator*

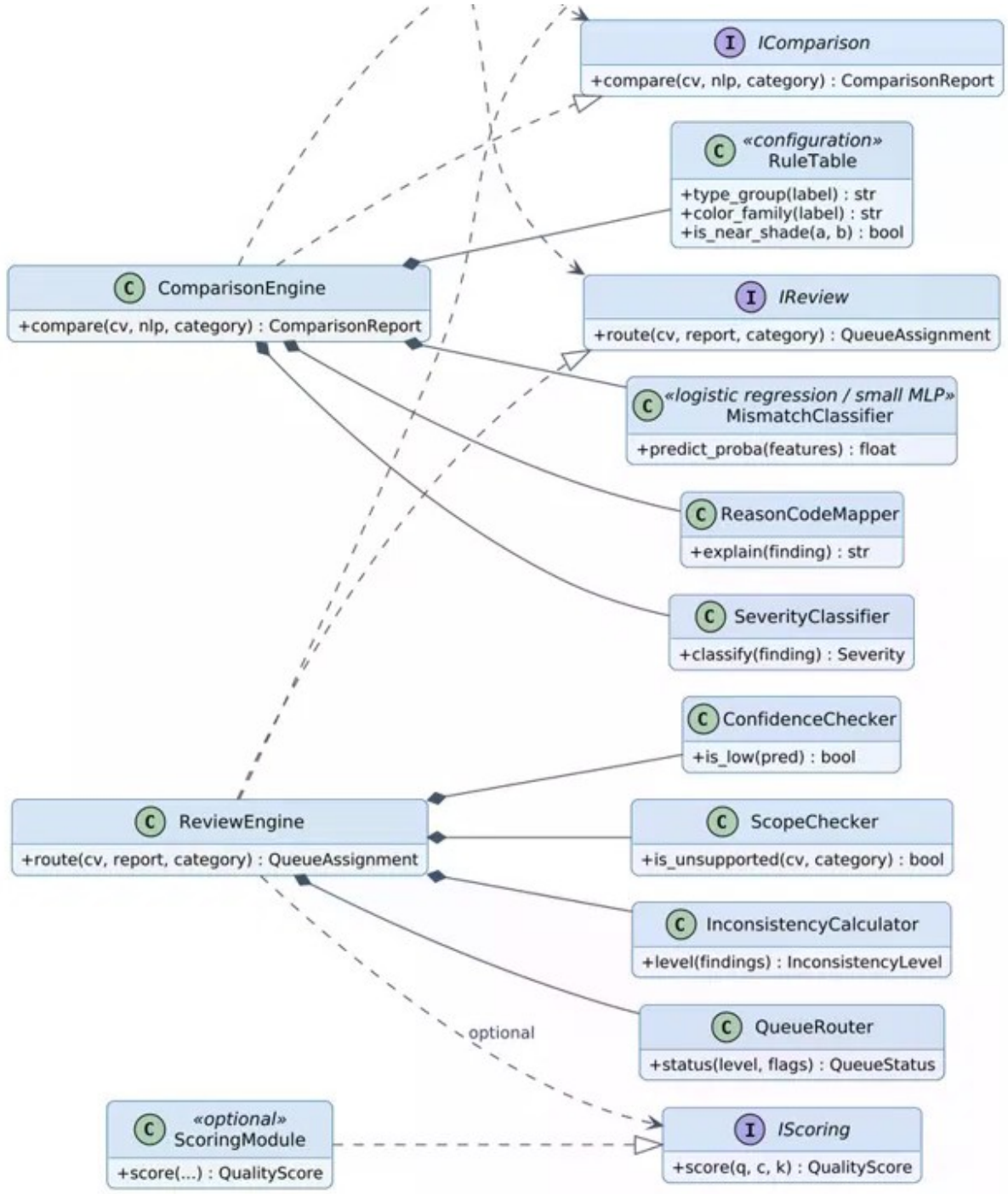


Figure 2. *Class diagram, design view (part 2 of 2): comparison, review, template and scoring components*

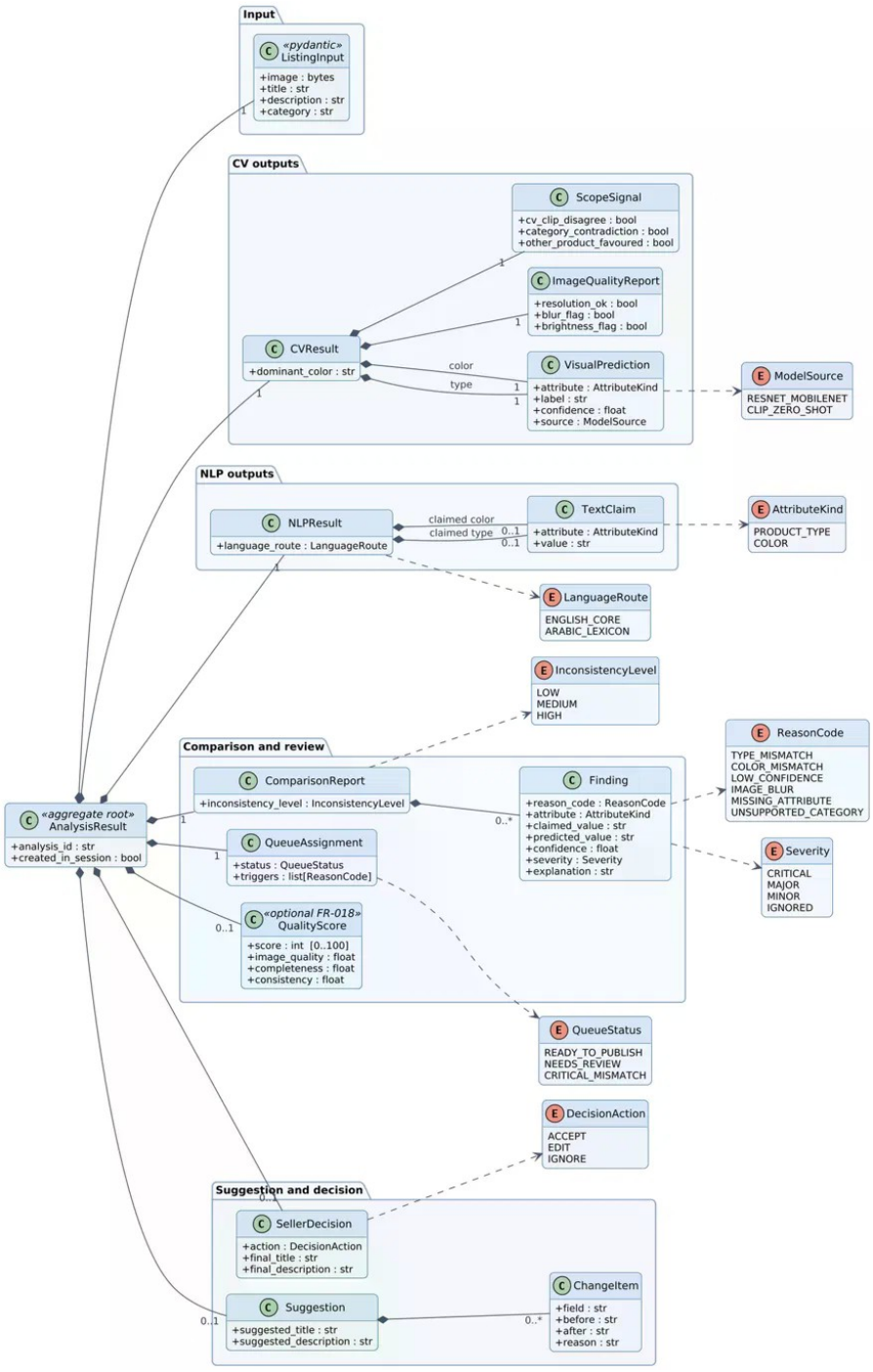


Figure 3. *Class diagram, data model: AnalysisResult aggregate and enumerations*

How to read the diagram

- Filled diamonds show composition: a module owns its helper classes (for example CVModule owns ImageQualityAnalyzer, DominantColorExtractor, TypeColorClassifier and ClipZeroShot).
- Dashed arrows with a hollow head mean "realises an interface" (ICvAnalysis, INlpExtraction, IComparison, IReview, ITemplateGeneration, IScoring); plain dashed arrows mean "uses".
- Stereotypes mark roles: «ResNet / MobileNet», «baseline / fallback», «configuration», «logistic regression / small MLP», «FastAPI», «optional» and «files».
- In the data model, AnalysisResult is the aggregate root. It owns the input listing, the CV and NLP results, the ComparisonReport with its findings, the QueueAssignment, an optional QualityScore, an optional Suggestion with change items and an optional SellerDecision.

Interfaces

Interface	Operation	Implemented by	Requirement
ICvAnalysis	analyze(image) : CVResult	CVModule	FR-004 to FR-006, FR-013
INlpExtraction	extract(title, description) : NLPResult	NLPModule	FR-007, FR-008
IComparison	compare(cv, nlp, category) : ComparisonReport	ComparisonEngine	FR-009 to FR-011, FR-016, FR-017
IReview	route(cv, report, category) : QueueAssignment	ReviewEngine	FR-012, FR-014, FR-015, FR-019, FR-020
ITemplateGeneration	suggest(cv, nlp, assignment) : Suggestion	TemplateGenerator	FR-021
IScoring (optional)	score(q, c, k) : QualityScore	ScoringModule	FR-018

Enumerations of the data model

Enumeration	Values
AttributeKind	PRODUCT_TYPE, COLOR
ModelSource	RESNET_MOBILENET, CLIP_ZERO_SHOT
LanguageRoute	ENGLISH_CORE, ARABIC_LEXICON
ReasonCode	TYPE_MISMATCH, COLOR_MISMATCH, LOW_CONFIDENCE, IMAGE_BLUR, MISSING_ATTRIBUTE, UNSUPPORTED_CATEGORY
Severity	CRITICAL, MAJOR, MINOR, IGNORED
InconsistencyLevel	LOW, MEDIUM, HIGH
QueueStatus	READY_TO_PUBLISH, NEEDS_REVIEW, CRITICAL_MISMATCH
DecisionAction	ACCEPT, EDIT, IGNORE

Design notes and traceability

Requirements	FR-004 to FR-025, NFR-MNT-01, NFR-MNT-02, NFR-TEC-04
Objectives	OBJ-01 to OBJ-08
Status	Reused from the earlier design work and re-checked against FR and NFR text
Open items	The session store is shown as in-memory with the label "D-07 TBC"; the register of section 11.3 has no D-07, so the label stands for the open item "duration and representation of seller decisions" (FRS 12.4). QueueStatus has no value for the Medium-only case until Q-01 is decided.

8.2 Object Diagram

The object diagram freezes one moment of one analysis to show how the classes of the data model are instantiated. It uses a listing whose title says "Red" while the image shows a black T-shirt, a different-family color contradiction.

Object diagram: one analysed listing (illustrative snapshot, values are examples, not measured results)

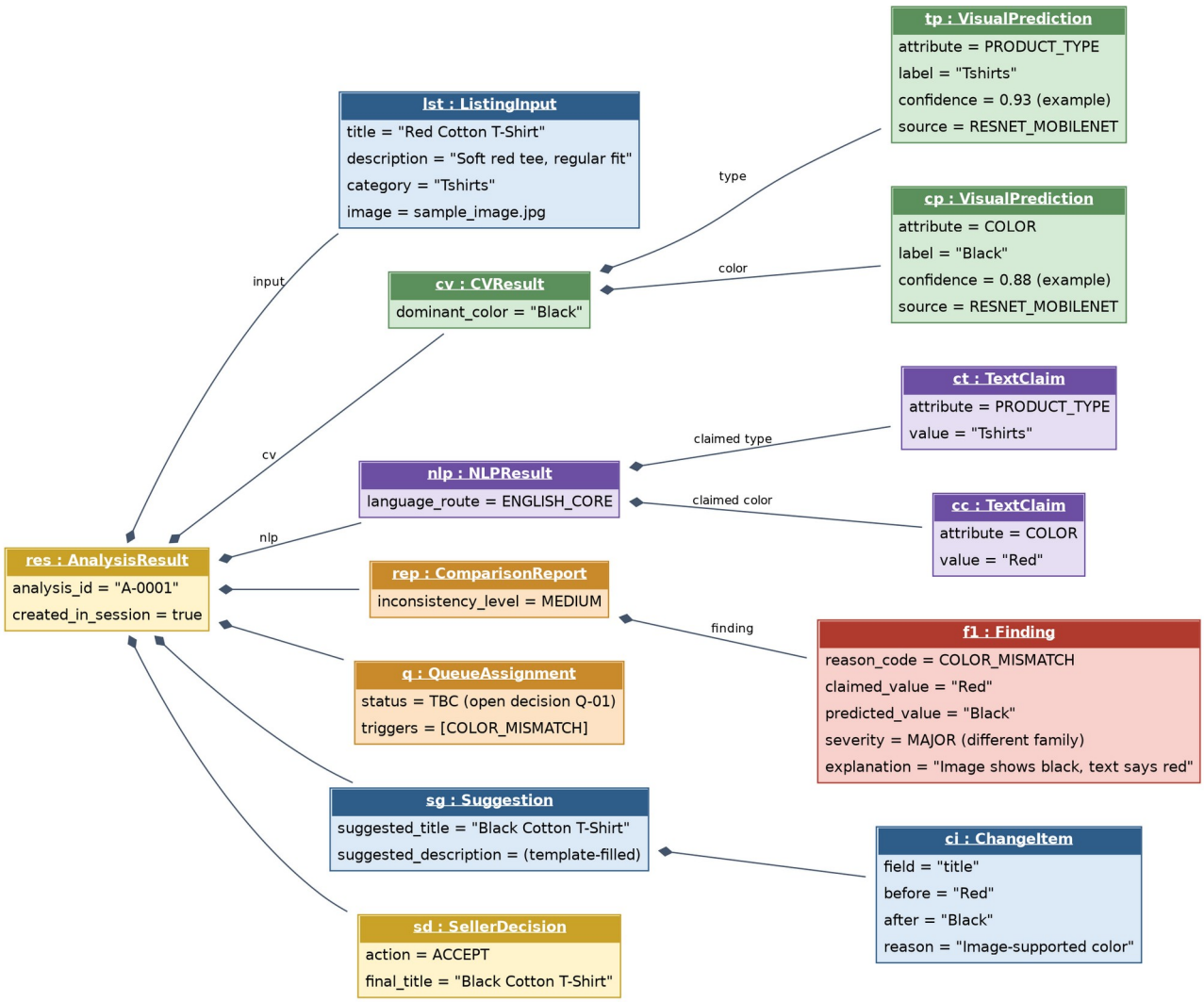


Figure 4. *Object diagram: one analysed listing (illustrative values)*

How to read the diagram

- Each box is an instance written name : Class and underlined; the lines are composition links from the AnalysisResult root.
- The type claim and prediction agree ("Tshirts"), so no TYPE_MISMATCH finding exists; the color claim "Red" contradicts the prediction "Black", so one COLOR_MISMATCH finding of severity MAJOR is created.
- One Major mismatch gives Inconsistency Level MEDIUM, and the QueueAssignment shows status "TBC (open decision Q-01)", exactly as the requirements leave it.
- The suggestion changes only the color word and the seller accepted it; no object represents publishing.

Design notes and traceability

Requirements	FR-009, FR-011, FR-016, FR-017, FR-019, FR-020 to FR-023
Objectives	OBJ-04, OBJ-05, OBJ-06, OBJ-08
Illustrative values	Confidence 0.93 and 0.88, the file name and the class "Tshirts" are examples; the five classes are not yet frozen (D-01).

8.3 Component Diagram

The component diagram shows the executable building blocks, the interfaces they provide and require, and the resources they read.

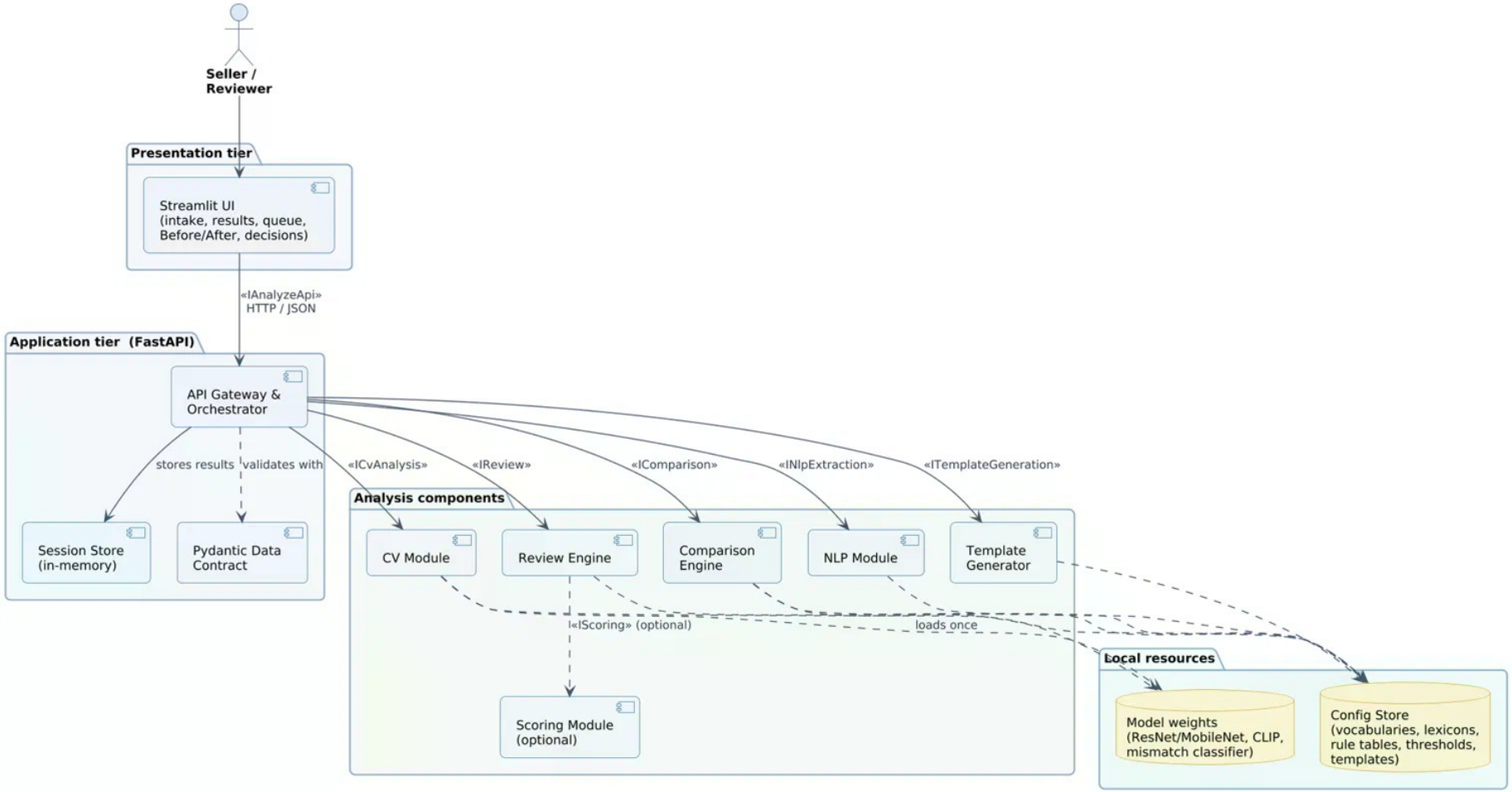


Figure 5. *Component diagram: tiers, components and interfaces*

How to read the diagram

- Packages group components by tier: presentation, application, analysis components and local resources.
- The orchestrator ("API Gateway and Orchestrator") requires ICvAnalysis, INlpExtraction, IComparison, IReview and ITemplateGeneration, and uses the optional IScoring through the review engine.
- The UI reaches the application tier only through the HTTP and JSON interface IAnalyzeApi.
- Analysis components read the Config Store and the Model weights store; weights are loaded once at start-up (NFR-PERF-08).

Design notes and traceability

Requirements	NFR-MNT-01 to NFR-MNT-03, NFR-TEC-01 to NFR-TEC-04
Objectives	OBJ-01
Design rule	Swapping a stub for a real module must not change callers (TAC-18).

8.4 Deployment Diagram

The deployment diagram maps the software to the single local workstation used for the demonstration.

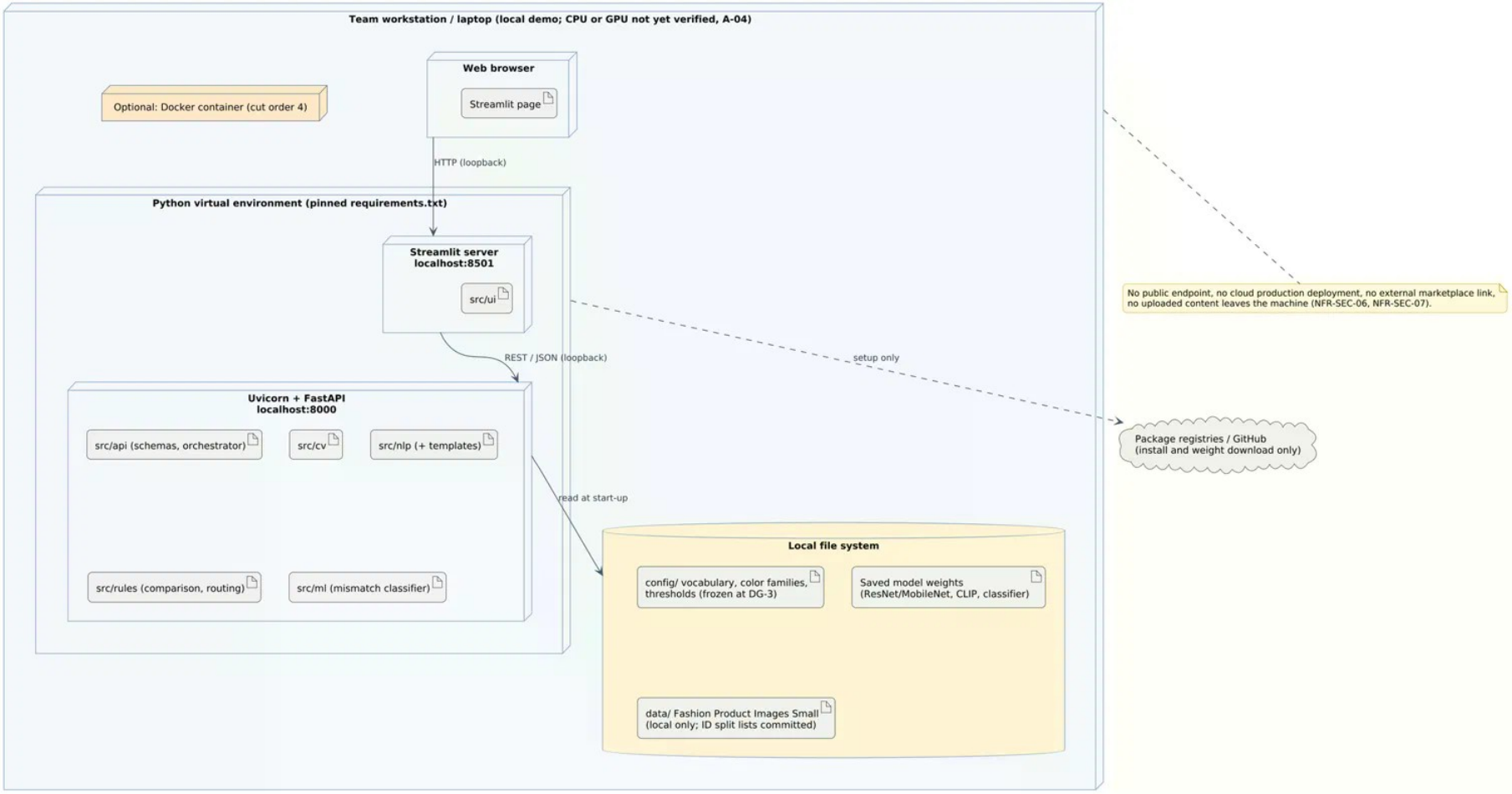


Figure 6. *Deployment diagram: local demo environment*

How to read the diagram

- The browser talks to the Streamlit server on localhost:8501 over loopback HTTP; Streamlit calls Uvicorn with FastAPI on localhost:8000 over loopback REST and JSON.
- Source packages src/api, src/cv, src/nlp, src/rules and src/ml run inside a Python virtual environment with pinned requirements (NFR-SEC-08).
- The local file system holds config/, saved model weights and the dataset with committed ID split lists; the cloud node is used only to install packages and download weights.
- Docker is drawn as optional (cut order 4). Hardware (CPU or GPU) is not yet verified (A-04, LIM-08).

Design notes and traceability

Requirements	NFR-SEC-06, NFR-SEC-07, NFR-TEC-15, CON-08
Objectives	OBJ-01, OBJ-09
Security note	No public endpoint, no cloud production deployment, no external marketplace link, and no uploaded content leaves the machine.

8.5 Composite Structure Diagram

The composite structure diagram opens the Analysis Orchestrator and shows its internal parts, ports and connectors.

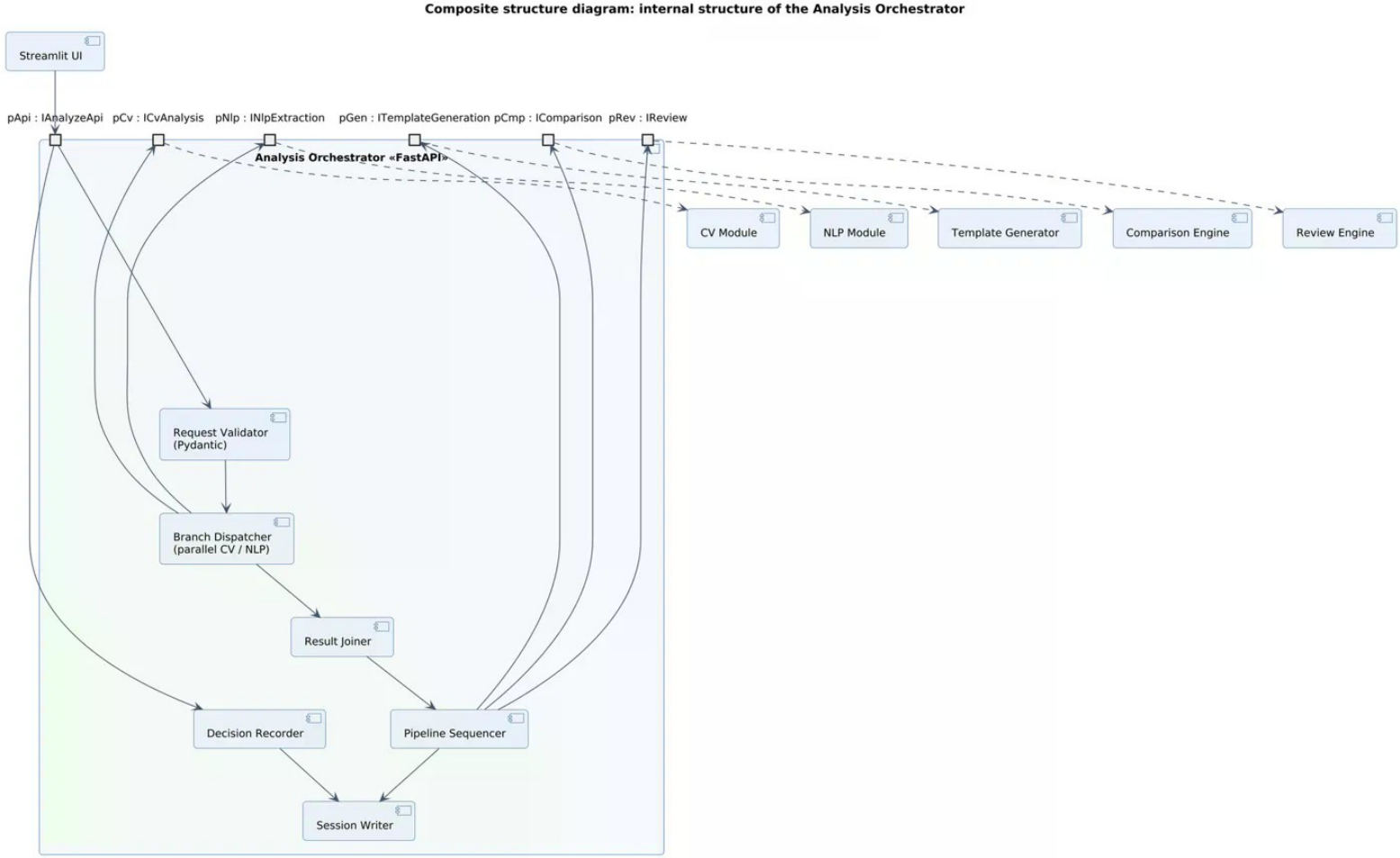


Figure 7. *Composite structure diagram: internal structure of the Analysis Orchestrator*

How to read the diagram

- Ports on the boundary (pApi, pCv, pNlp, pGen, pCmp, pRev) expose the interfaces the orchestrator provides or requires.
- A request enters through pApi to the Request Validator (Pydantic), then to the Branch Dispatcher, which calls the CV and NLP ports in parallel.
- The Result Joiner passes combined results to the Pipeline Sequencer, which calls comparison, review and template generation and hands the result to the Session Writer.
- The Decision Recorder receives seller decisions from pApi and writes them through the Session Writer.

Design notes and traceability

Requirements	FR-001 to FR-003, FR-023 to FR-025, NFR-PERF-02, NFR-REL-04
Objectives	OBJ-01, OBJ-08

8.6 Package Diagram

The package diagram organises the repository into packages and shows their dependencies, so that work packages can be divided without circular dependencies.

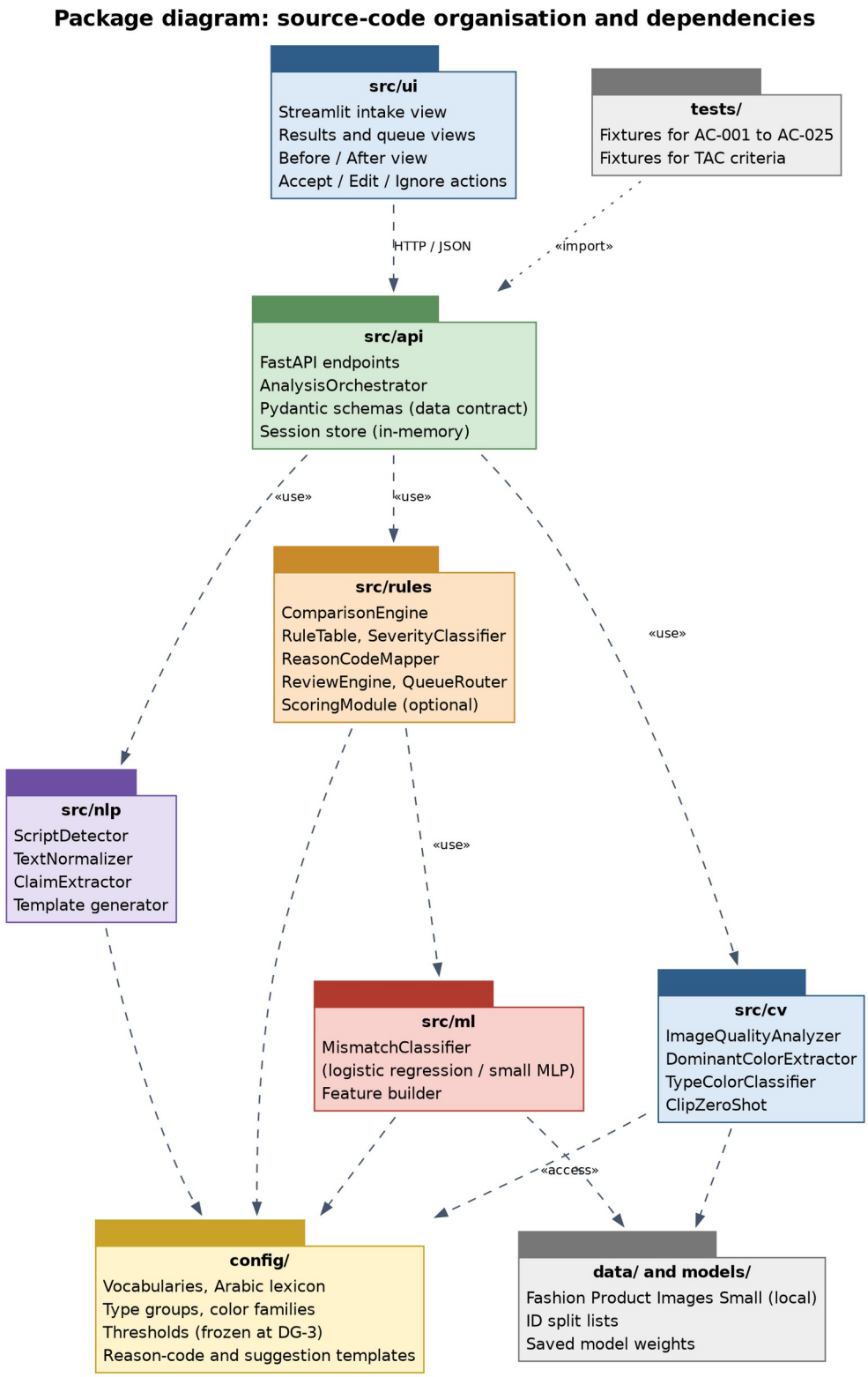


Figure 8. *Package diagram: source-code organisation and dependencies*

How to read the diagram

- src/ui depends on src/api over HTTP and JSON; src/api uses src/cv, src/nlp and src/rules; src/rules uses src/ml for the mismatch classifier.
- src/cv, src/nlp, src/rules and src/ml access config/ for vocabularies, rule tables, thresholds and templates; src/cv and src/ml also access data/ and model weights.
- Template generation lives with src/nlp; the Pydantic schemas live in src/api so that every package shares one contract.
- tests/ holds fixtures for AC-001 to AC-025 and the TAC criteria and imports the packages under test.

Package responsibilities and owners

Package	Contents	Lead role (WBS)
src/ui	Streamlit intake, results, queue, Before / After, decisions	FQA (3.14, 3.15)
src/api	FastAPI endpoints, orchestrator, schemas, session store	PMI (3.13)
src/cv	Quality checks, type and color model, dominant color, CLIP	DL/CV (3.2 to 3.4)
src/nlp	Script detection, normalisation, claim extraction, templates	NLP (3.6, 3.11)
src/rules	Comparison, severity, reason codes, review routing, score	RDL (3.9, 3.10)
src/ml	Mismatch classifier and features	ML (3.7, 3.8)
config/	Vocabularies, groups, families, thresholds, templates	RDL (2.3), ML (3.5)
tests/	Fixtures and tests	FQA (2.10, 4.1)

Design notes and traceability

Requirements	NFR-MNT-01, NFR-MNT-04, NFR-MNT-09, NFR-TEC-01
Objectives	OBJ-01, OBJ-09
Roles	Technical roles come from the Project Management document [3]: DL/CV, ML, NLP, RDL, FQA and PMI.

8.7 Profile Diagram

The profile diagram defines the small UML profile used in this project. Stereotypes extend UML metaclasses and carry tagged values so that models can express project-specific roles consistently.

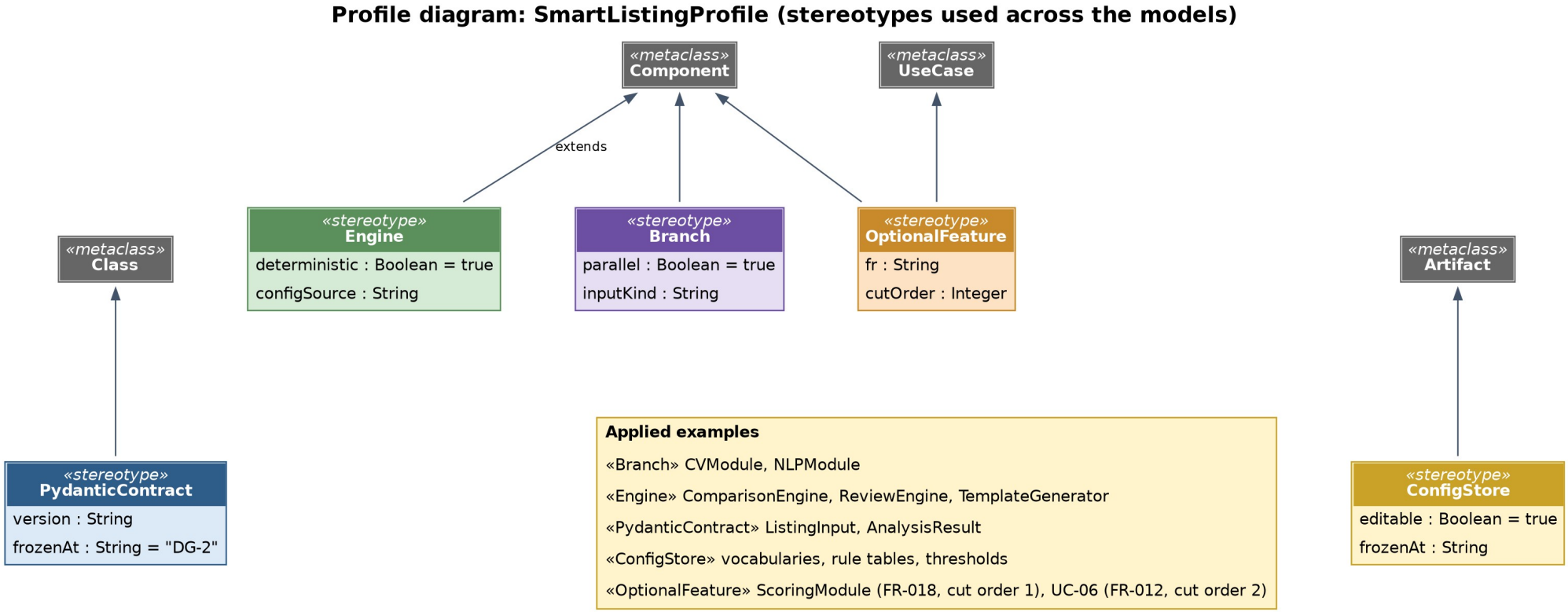


Figure 9. *Profile diagram: SmartListingProfile*

How to read the diagram

- «Engine» extends Component and records that the component is deterministic and which configuration drives it.
- «Branch» extends Component for the parallel CV and NLP branches.
- «PydanticContract» extends Class and records the contract version and the gate at which it is frozen (DG-2).
- «ConfigStore» extends Artifact for editable, gate-frozen configuration; «OptionalFeature» extends UseCase and Component and records the FR and the cut order.

Design notes and traceability

Requirements	NFR-REL-03 (deterministic decisions), NFR-MNT-04, NFR-MNT-06
Objectives	OBJ-01
Use	The stereotypes label the class, component and use case diagrams.

9. UML Behavioral Diagrams

Behavioral diagrams describe how the system acts: who uses it, how work flows, how a listing changes state, how objects exchange messages and how events are ordered in time.

9.1 Use Case Diagram

The use case diagram shows the two human actors, the system boundary and the thirteen use cases with their include, extend and generalisation relationships.

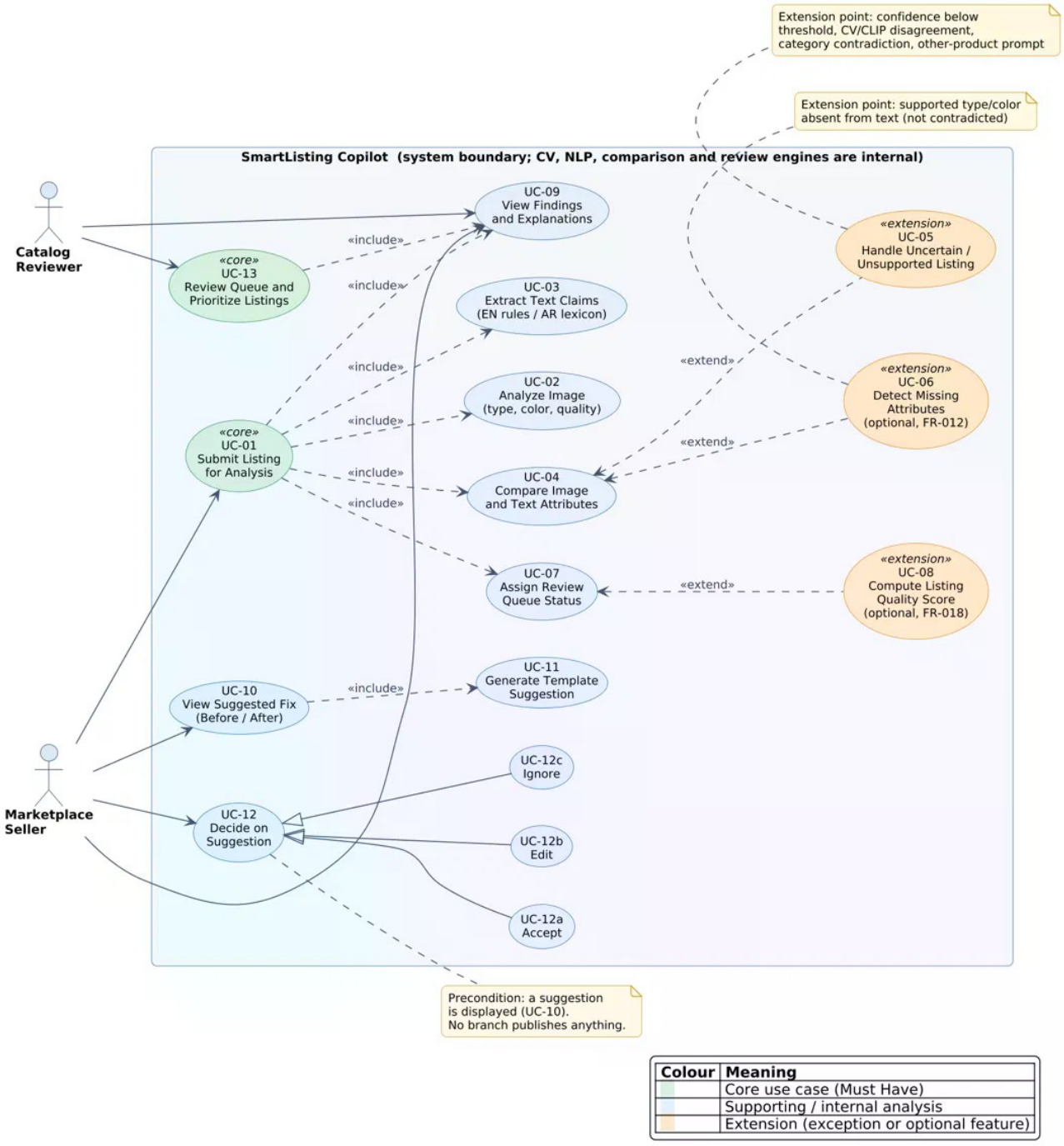


Figure 10. *Use case diagram: SmartListing Copilot*

How to read the diagram

- The Marketplace Seller starts UC-01, views findings (UC-09), views the suggested fix (UC-10) and decides (UC-12). The Catalog Reviewer reviews the queue (UC-13) and views findings (UC-09).

- Dashed «include» arrows show mandatory sub-behavior; for example UC-01 includes UC-02, UC-03, UC-04, UC-07 and UC-09, and UC-10 includes UC-11.
- Orange «extension» use cases (UC-05, UC-06, UC-08) extend UC-04 or UC-07 only when their trigger or feature flag applies.
- UC-12 is generalised into Accept (12a), Edit (12b) and Ignore (12c). Internal modules are inside the boundary and are not actors.

Actors

Actor	Type	Description
Marketplace Seller	Primary	Submits a listing, reads findings and the suggestion, and accepts, edits or ignores it
Catalog Reviewer	Primary (review user)	Opens the queue, inspects flagged listings, reads reason codes and evidence
Academic Supervisor	Not a system actor	Reviews documents only
Operations / Support	Indirect beneficiary	No system role

Actor-to-use-case matrix

Use case	Seller	Reviewer
UC-01 Submit Listing for Analysis	Yes	
UC-02 to UC-08 (internal analysis)	via UC-01	via UC-13
UC-09 View Findings and Explanations	Yes	Yes
UC-10 View Suggested Fix	Yes	
UC-11 Generate Template Suggestion	via UC-10	
UC-12 Decide on Suggestion (12a, 12b, 12c)	Yes	
UC-13 Review Queue and Prioritize Listings		Yes

Use case relationships

From	Relationship	To	Condition
UC-01	include	UC-02, UC-03, UC-04, UC-07, UC-09	Always
UC-05	extend	UC-04	Low confidence, CV and CLIP disagreement, category contradiction, other-product prompt
UC-06	extend	UC-04	Supported type or color absent from text (FR-012 enabled)
UC-08	extend	UC-07	Quality Score enabled (FR-018)
UC-10	include	UC-11	Always
UC-13	include	UC-09	Always
UC-12a, 12b, 12c	generalisation	UC-12	Accept, Edit or Ignore

Design notes and traceability

Requirements	FR-001 to FR-025 (see traceability in section 11.1)
Objectives	OBJ-01 to OBJ-08
Status	Reused from the requirements document and the earlier design work; consistent with Figure 1 of the requirements document [4]

9.1.1 Use Case Descriptions

Each description gives actor, goal, preconditions, trigger, basic flow, alternative flows, postconditions, business rules, open items and acceptance links. Content follows the requirements document [4]; open items are not resolved by assumption.

UC-01 Submit Listing for Analysis

Actor	Marketplace Seller
Classification	Core (Must Have)
Related FR / OBJ / US	FR-001, FR-002, FR-003 / OBJ-01 / US-01
Goal	Obtain a listing-quality assessment before publication.
Preconditions	The Streamlit intake view is open and the FastAPI backend is running.
Trigger	The seller presses Analyze.
Basic flow	1. The seller provides one image, a title, a description and a category. 2. The system validates the input against the Pydantic contract and checks that a readable image is present. 3. The system passes the image to the CV branch, the text to the NLP branch and the category to the scope check. 4. The system runs UC-02 and UC-03 in parallel. 5. The system continues with UC-04, UC-05 or UC-06 where triggered, and UC-07. 6. The system presents the results (UC-09) and, where available, the suggestion (UC-10).
Alternative flows	A1 Missing image: analysis does not start and a non-technical error is shown (confirmed behavior). A2 Unreadable, corrupt or unsupported image: normal analysis cannot proceed; how the error is presented is open (IN-03). A3 Empty title or description: reject, or treat as absent claims, is open (IN-01, IN-02). A4 Blank or malformed category: open (IN-04).
Postconditions	Findings and a queue status are available. Nothing is published.
Business rules	Exactly one listing per request. Title, description and category are data, never executed (NFR-SEC-03).
Open items	IN-01 to IN-04
Acceptance	AC-001 to AC-003, TAC-UC01

UC-02 Analyze Image (internal)

Actor	CV module (system)
Classification	Supporting
Related FR / OBJ / US	FR-004, FR-005, FR-006, FR-013 / OBJ-02, OBJ-01 / US-02
Goal	Produce structured visual attributes, confidence and quality findings from the image.
Preconditions	A readable image has been accepted by UC-01.
Trigger	Included by UC-01 after validation.
Basic flow	1. Preprocess the image. 2. Check resolution, blur and brightness. 3. Predict Product Type with ResNet or MobileNet (CLIP as baseline or fallback). 4. Predict Color and cross-check it with the dominant color. 5. Return predictions with confidence, quality flags and scope signals.
Alternative flows	A1 Blur rule triggered: IMAGE_BLUR flag. A2 Confidence below threshold: LOW_CONFIDENCE (see UC-05). A3 Out-of-scope image: no forced label (see UC-05).

	A4 Primary model unavailable or in disagreement beyond the scope rule: use CLIP zero-shot or escalate to review (NFR-REL-02).
Postconditions	Structured visual attributes, confidence and quality findings are available.
Business rules	Confidence is never described as proof of correctness (FR-013). Model weights are loaded once at start-up.
Open items	D-01, D-03
Acceptance	AC-004 to AC-006, AC-013

UC-03 Extract Text Claims (internal)

Actor	NLP module (system)
Classification	Supporting
Related FR / OBJ / US	FR-007, FR-008 / OBJ-03 / US-03
Goal	Extract only the claimed Product Type and Color from title and description.
Preconditions	Title and description text has been submitted.
Trigger	Included by UC-01.
Basic flow	1. Detect Arabic characters. 2. Select the fixed Arabic lexicon or the English core vocabulary. 3. Normalise the text. 4. Extract the type claim and color claim using rules and fuzzy matching. 5. Return the claims and the language route.
Alternative flows	A1 No claim found: the field stays empty and nothing is inferred (feeds UC-06). A2 Mixed script: Arabic-script input routes to the Arabic lexicon (FR-008).
Postconditions	Normalised claimed type and claimed color are available.
Business rules	Only type and color are extracted; no size, material or price. Arabic coverage is deliberately narrow (LIM-04).
Open items	None structural
Acceptance	AC-007, AC-008

UC-04 Compare Image and Text Attributes (internal)

Actor	Comparison engine (system)
Classification	Supporting
Related FR / OBJ / US	FR-009, FR-010, FR-011, FR-013, FR-016, FR-017, FR-019 / OBJ-04, OBJ-05 / US-04, US-06
Goal	Detect contradictions and attach severity, reason codes and explanations.
Preconditions	CV predictions and NLP claims exist.
Trigger	Included by UC-01 after UC-02 and UC-03 complete.
Basic flow	1. Compare predicted and claimed Product Type. 2. Compare predicted and claimed Color using family and near-shade rules. 3. Assign severity: cross-group type = Critical; same-group type = Major; different-family color = Major; near shade = Minor or ignored. 4. Attach a reason code, confidence and a plain-language explanation to each finding. 5. Compute the Listing Inconsistency Level (Low, Medium, High).
Alternative flows	A1 Consistent attributes: no finding. A2 Claim absent: see UC-06.

	A3 Uncertain or unsupported input: see UC-05.
Postconditions	A mismatch report and an inconsistency level exist.
Business rules	Rules are authoritative; the lightweight classifier supports but does not override them (NFR-TEC-10). Size, material and price are never compared.
Open items	D-01, D-02
Acceptance	AC-009 to AC-011, AC-016, AC-017, AC-019, TAC-UC04

UC-05 Handle Uncertain / Unsupported Listing (extends UC-04)

Actor	Review engine (system)
Classification	Extension
Related FR / OBJ / US	FR-014, FR-015 / OBJ-05, OBJ-06 / US-05
Goal	Never present an uncertain or out-of-scope listing as a confident result.
Preconditions	UC-04 is running.
Trigger	Confidence is below threshold, or CV and CLIP disagree, or the predicted type contradicts the seller category, or CLIP favors the other-product prompt.
Basic flow	1. Raise LOW_CONFIDENCE and / or UNSUPPORTED_CATEGORY. 2. Avoid a forced definitive type. 3. Route the listing to Needs review. 4. Show the plain wording "Please verify".
Alternative flows	A1 Several triggers at once: one warning per reason code per attribute is shown, most severe first (NFR-USA-09).
Postconditions	The listing is never confidently auto-accepted.
Business rules	Threshold is validation-derived and frozen at DG-3 (D-03).
Open items	D-03; Q-02 when combined with a Critical mismatch
Acceptance	AC-014, AC-015, TAC-UC05

UC-06 Detect Missing Attributes (extends UC-04, optional)

Actor	Comparison engine (system)
Classification	Extension (Should Have, cut order 2)
Related FR / OBJ / US	FR-012 / OBJ-06 / US-05
Goal	Flag a supported type or color that the text does not mention.
Preconditions	UC-03 returned an empty claim while UC-02 supports the attribute.
Trigger	A visually supported type or color is absent from the text.
Basic flow	1. Detect that the supported attribute is absent from the claims. 2. Raise MISSING_ATTRIBUTE (type or color). 3. Route the listing to Needs review.
Alternative flows	A1 Contradicted claim: treated as a mismatch (UC-04), not as missing. A2 Feature disabled under the fallback: the core workflow still works (TAC-17).
Postconditions	A missing-attribute flag exists when the feature is enabled.
Business rules	Can be cut under the proposal fallback without affecting FR-006 findings.
Open items	IN-02 (empty description) may adjust this case

Acceptance	AC-012

UC-07 Assign Review Queue Status (internal)

Actor	Review engine (system)
Classification	Supporting
Related FR / OBJ / US	FR-020 / OBJ-06 / US-06
Goal	Give every analysed listing one of the three queue states.
Preconditions	Findings, inconsistency level and review flags exist.
Trigger	Included by UC-01.
Basic flow	1. A clean listing at Low level goes to Ready to publish. 2. Any review flag (low confidence, blur, missing attribute, unsupported category) goes to Needs review. 3. High inconsistency or any Critical finding goes to Critical mismatch.
Alternative flows	A1 Exactly one Major (Medium level) with no other flag: queue state undecided (Q-01); the model shows "Pending decision (TBC)". A2 Conflicting triggers such as Critical plus low confidence: precedence undecided (Q-02).
Postconditions	A queue status is assigned. It is a local decision-support label only.
Business rules	No default is assumed for Q-01 or Q-02.
Open items	Q-01, Q-02
Acceptance	AC-020, TAC-UC07

UC-08 Compute Listing Quality Score (extends UC-07, optional)

Actor	Scoring module (system)
Classification	Extension (Should Have, cut order 1)
Related FR / OBJ / US	FR-018 / OBJ-01 / US-06
Goal	Provide a transparent 0 to 100 heuristic for reviewer attention.
Preconditions	Image quality, completeness and consistency inputs exist and the feature is enabled.
Trigger	Quality Score enabled.
Basic flow	1. Combine image quality, attribute completeness and image-text consistency. 2. Produce a score between 0 and 100. 3. Attach it to the AnalysisResult.
Alternative flows	A1 Feature disabled: the score is omitted; FR-006 still reports image-quality findings.
Postconditions	A heuristic score exists when enabled.
Business rules	Model confidence and business-return predictions are excluded.
Open items	D-05 (weights)
Acceptance	AC-018

UC-09 View Findings and Explanations

Actor	Marketplace Seller, Catalog Reviewer
Classification	Supporting
Related FR / OBJ / US	FR-013, FR-016, FR-017 / OBJ-05 / US-02, US-04

Goal	See the evidence behind every warning.
Preconditions	An analysis result exists.
Trigger	The actor opens the results.
Basic flow	1. The actor opens the results. 2. For each warning the system shows claimed value, predicted value, confidence, severity, reason code and plain-language rationale. 3. The system shows the queue status and the supporting predictions.
Alternative flows	A1 No findings: the system shows "Ready to publish" with the supporting predictions.
Postconditions	Evidence is visible. Confidence is not presented as correctness.
Business rules	Severity and queue status are distinguished by text labels, not color alone (NFR-USA-03).
Open items	None
Acceptance	AC-013, AC-016, TAC-UC09

UC-10 View Suggested Fix (Before / After)

Actor	Marketplace Seller
Classification	Supporting
Related FR / OBJ / US	FR-021, FR-022 / OBJ-07 / US-07
Goal	Compare a proposed listing with the original before deciding.
Preconditions	At least one finding supports a correction.
Trigger	The seller opens the suggestion view.
Basic flow	1. The system generates a template suggestion (UC-11). 2. The system shows original and suggested text side by side, with changes highlighted and reasons stated.
Alternative flows	A1 Uncertain or unsupported listing: the suggestion uses only model-supported attributes, or the system shows no suggestion for an unsupported type.
Postconditions	The seller can compare before deciding. Nothing is applied.
Business rules	No size, material or price claim is ever added.
Open items	None
Acceptance	AC-021, AC-022, TAC-UC10

UC-11 Generate Template Suggestion (internal)

Actor	Template generator (system)
Classification	Supporting
Related FR / OBJ / US	FR-021 / OBJ-07 / US-07
Goal	Produce a bounded corrected title and description.
Preconditions	Supported, confident type or color predictions exist.
Trigger	Included by UC-10.
Basic flow	1. Select the controlled title and description templates. 2. Fill them using only the predicted Product Type and Color. 3. Return the suggestion and the change items (field, before, after, reason).
Alternative flows	A1 No supported and confident type or color: return "no definitive suggestion" (Please verify).

Postconditions	A suggestion or an explicit no-suggestion result exists.
Business rules	No LLM; every generated attribute is traceable to a prediction.
Open items	None
Acceptance	AC-021

UC-12 Decide on Suggestion (Accept / Edit / Ignore)

Actor	Marketplace Seller
Classification	Core, generalised into 12a, 12b, 12c
Related FR / OBJ / US	FR-023, FR-024, FR-025 / OBJ-08 / US-08
Goal	Keep final control over listing wording.
Preconditions	A suggestion is displayed (UC-10).
Trigger	The seller chooses one option.
Basic flow	1. 12a Accept: the correction is recorded as seller-approved in the current review flow. 2. 12b Edit: the seller changes the wording, confirms, and the revised wording is retained. 3. 12c Ignore: the original text is kept and the suggestion is left unapplied.
Alternative flows	A1 No decision: nothing is applied.
Postconditions	No external publish action is performed in any branch.
Business rules	Accept, Edit and Ignore are always visible together (NFR-USA-05).
Open items	Duration and representation of seller decisions (FRS 12.4)
Acceptance	AC-023 to AC-025, TAC-UC12

UC-13 Review Queue and Prioritize Listings

Actor	Catalog Reviewer
Classification	Core (Must Have)
Related FR / OBJ / US	FR-016, FR-017, FR-019, FR-020 / OBJ-06 / US-04, US-05, US-06
Goal	Decide which flagged listings to open first.
Preconditions	Analysed listings exist in the session.
Trigger	The reviewer opens the queue.
Basic flow	1. The reviewer opens the queue. 2. Listings appear grouped as Critical mismatch, Needs review and Ready to publish. 3. The reviewer opens a flagged listing and reads its reason codes and evidence (includes UC-09).
Alternative flows	A1 Empty queue: no listings are shown.
Postconditions	The reviewer knows where to look first.
Business rules	The MVP defines no reviewer accounts and no separate reviewer workflow beyond viewing the queue and findings.
Open items	Q-01, Q-02 affect grouping of Medium-only listings
Acceptance	AC-016, AC-017, AC-019, AC-020, TAC-UC13

Use case summary

UC	Name	Actor	Class	FRs	Objective
UC-01	Submit Listing for Analysis	Marketplace Seller	Core	001,002,003	OBJ-01
UC-02	Analyze Image	CV module	Supporting	004,005,006,013	OBJ-02
UC-03	Extract Text Claims	NLP module	Supporting	007,008	OBJ-03
UC-04	Compare Image and Text Attributes	Comparison engine	Supporting	009,010,011,013, 016,017,019	OBJ-04
UC-05	Handle Uncertain / Unsupported Listing	Review engine	Extension	014,015	OBJ-05
UC-06	Detect Missing Attributes	Comparison engine	Extension	012	OBJ-06
UC-07	Assign Review Queue Status	Review engine	Supporting	020	OBJ-06
UC-08	Compute Listing Quality Score	Scoring module	Extension	018	OBJ-01
UC-09	View Findings and Explanations	Marketplace Seller, Catalog Reviewer	Supporting	013,016,017	OBJ-05
UC-10	View Suggested Fix	Marketplace Seller	Supporting	021,022	OBJ-07
UC-11	Generate Template Suggestion	Template generator	Supporting	021	OBJ-07
UC-12	Decide on Suggestion	Marketplace Seller	Core	023,024,025	OBJ-08
UC-13	Review Queue and Prioritize Listings	Catalog Reviewer	Core	016,017,019,020	OBJ-06

9.2 Activity Diagram

The swimlane activity diagram follows one listing from the seller entering data to the seller deciding on a suggestion, and shows the parallel CV and NLP branches. The diagram is on the next page.

How to read the diagram

- Lanes (left to right): Seller, Streamlit UI, FastAPI Orchestrator, CV Module, NLP Module, Decision Engines.
- A fork after validation starts the CV branch (preprocess, quality checks, type, color, confidence and scope signals) and the NLP branch (script detection, lexicon choice, normalisation, claim extraction); a join waits for both.
- Decision Engines compare, assign severity and reason codes, raise flags, compute the Inconsistency Level, optionally compute the score, assign the queue status and either fill templates or show "Please verify".
- The seller finally chooses Accept, Edit or Ignore, or does nothing, in which case nothing is applied.

Decision nodes

Decision	Yes	No	Requirement
Image present and readable?	Start parallel branches	Show non-technical error; analysis does not start (IN-01 to IN-04 TBC)	FR-001
Arabic characters?	Fixed Arabic lexicon	English core vocabulary	FR-008
Quality Score enabled?	Compute score 0 to 100	Skip	FR-018
Supported and confident type / color?	Fill controlled templates	No definitive suggestion ("Please verify")	FR-021, FR-014
Seller decision?	Accept, Edit or Ignore as chosen	Nothing is applied	FR-023 to FR-025

Activity diagram: listing analysis workflow (swimlanes)

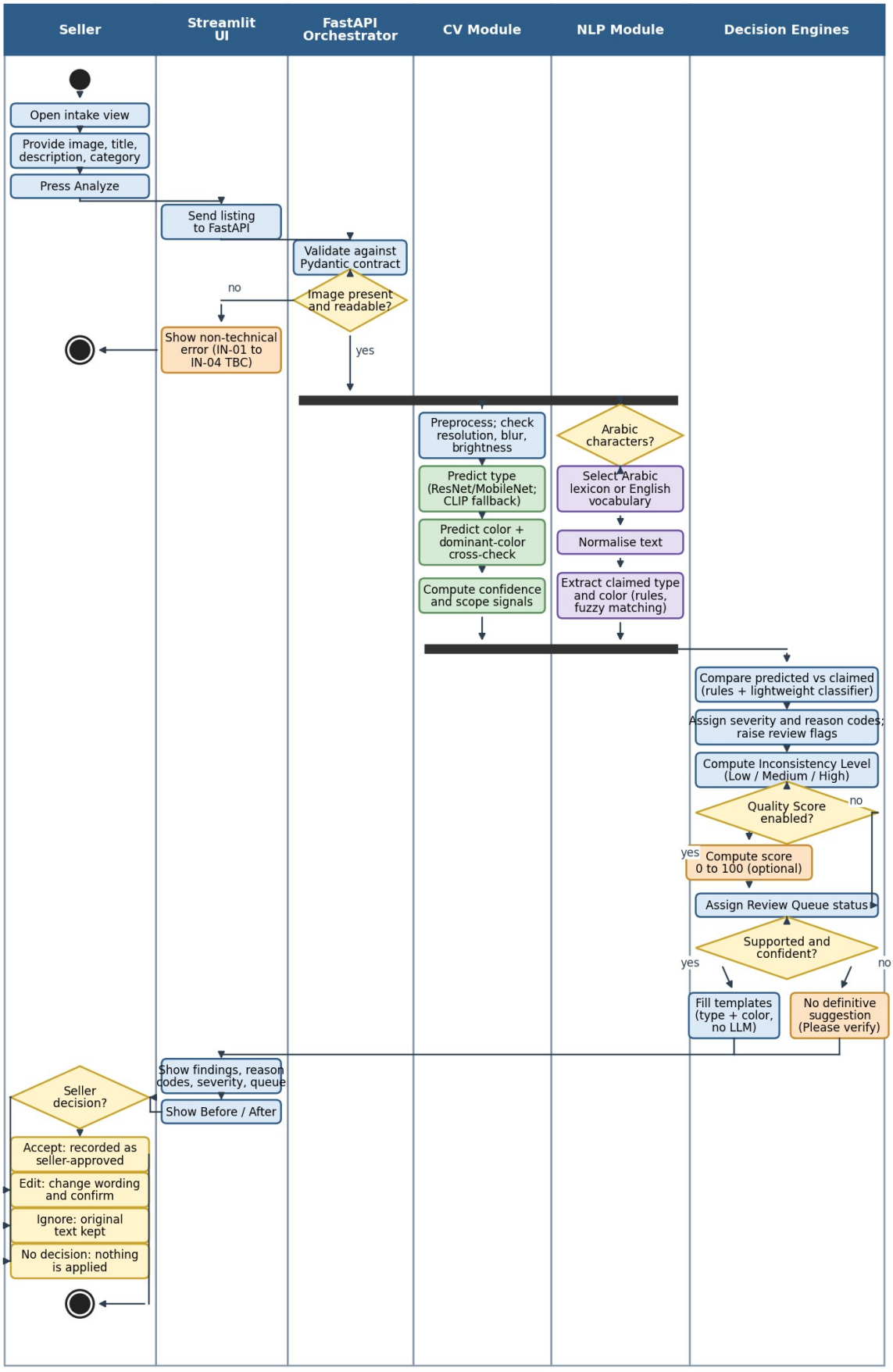


Figure 11. *Activity diagram: listing analysis workflow (swimlanes)*

Design notes and traceability

Requirements	FR-001 to FR-025 (main flow), NFR-PERF-02
Objectives	OBJ-01, OBJ-06, OBJ-07, OBJ-08
Open items	Empty-field and corrupt-file handling (IN-01 to IN-04); queue precedence (Q-01, Q-02)
Status	Redrawn for readability from the earlier activity diagram, same content

9.3 State Machine Diagram

The state machine shows the lifecycle of one analysed listing from the first open of the intake view to the closed session state.

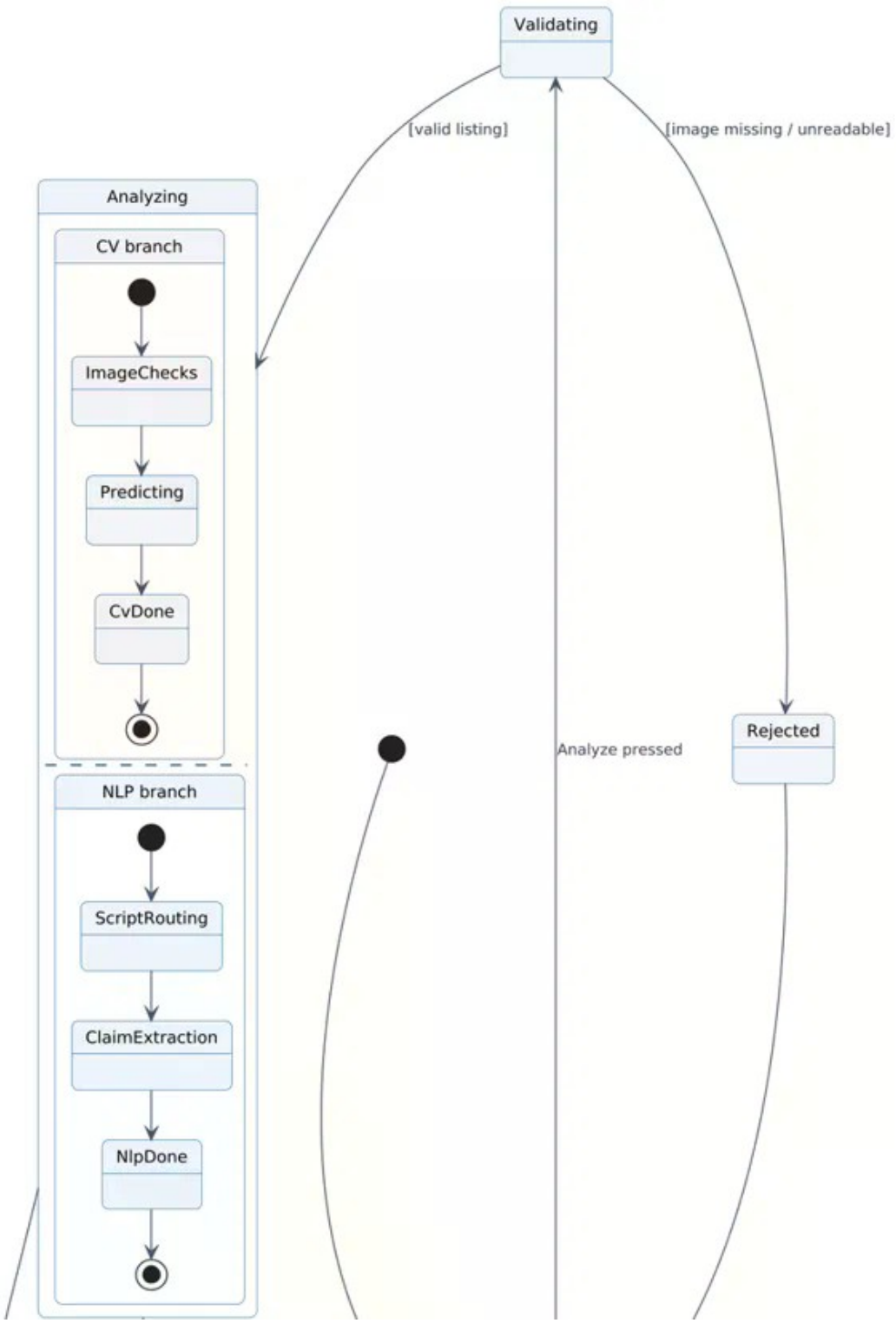


Figure 12. *State machine diagram (part 1 of 2): intake, validation and parallel analysis*

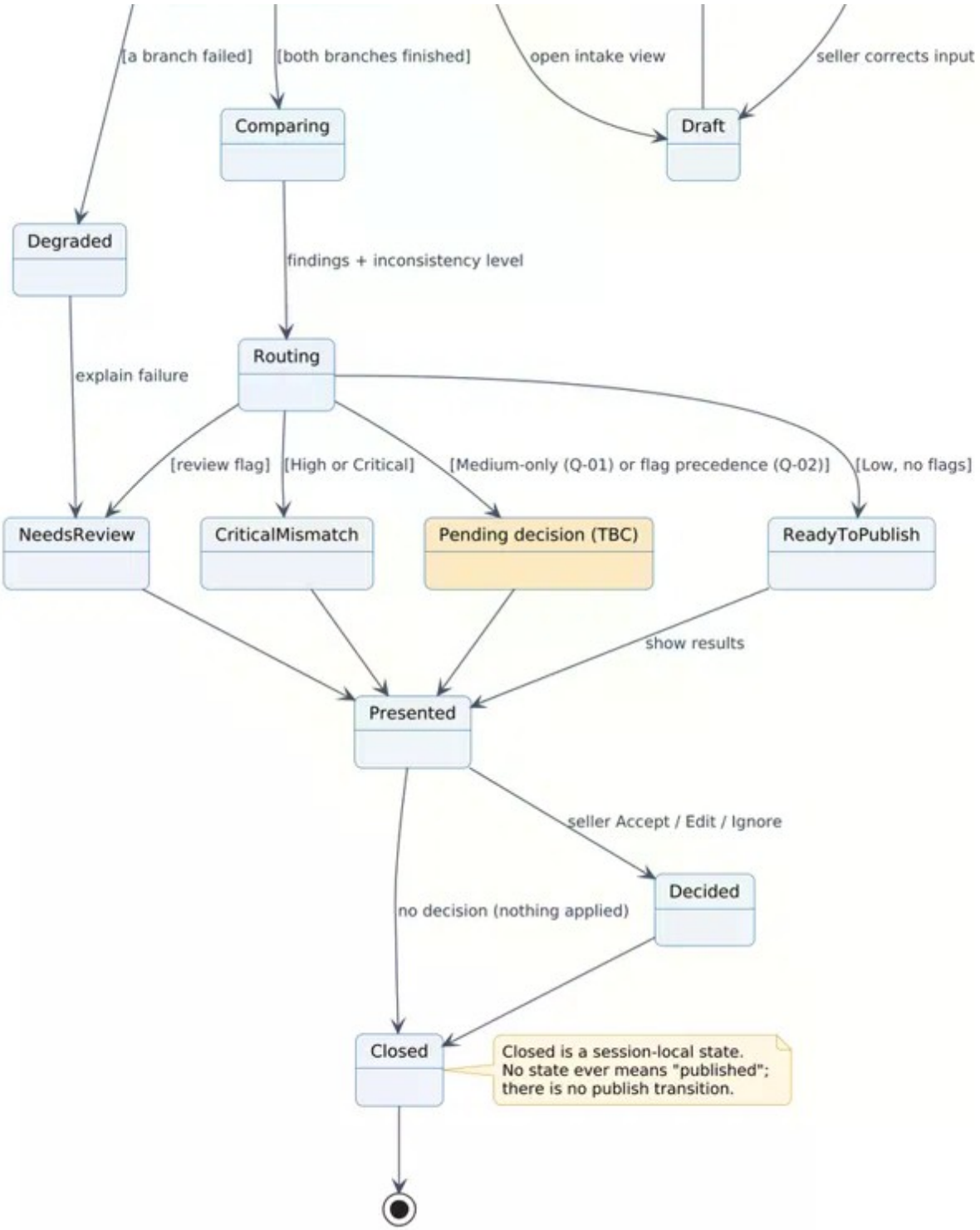


Figure 13. *State machine diagram (part 2 of 2): comparison, routing, presentation and decision*

How to read the diagram

- Draft leads to Validating when Analyze is pressed; an unreadable or missing image leads to Rejected and back to Draft after the seller corrects the input.
- Analyzing is a composite state with two concurrent regions (CV branch and NLP branch); both must finish before Comparing. A failure in a branch leads to Degraded and then to NeedsReview with an explanation.
- Routing enters one of NeedsReview, ReadyToPublish, CriticalMismatch or the orange "Pending decision (TBC)" state, then Presented.
- Presented leads to Decided when the seller accepts, edits or ignores, or directly to Closed when no decision is made. Closed is session-local; no state ever means "published".

State catalogue

State	Meaning	Leaves on
Draft	Intake view open, listing being entered	Analyze pressed
Validating	Pydantic and image checks run	Valid listing / image missing or unreadable
Rejected	Input refused; analysis not started	Seller corrects input
Analyzing	CV and NLP branches run concurrently	Both branches finished / a branch failed
Degraded	One branch failed; application keeps running	Failure is explained
Comparing	Type and color comparison, severity, reason codes	Findings and inconsistency level ready
Routing	Queue rules applied	Review flag / Low with no flags / High or Critical / Medium-only or flag precedence
NeedsReview	Flagged for human attention	Results shown
ReadyToPublish	Clean listing at Low level (label only)	Results shown
CriticalMismatch	High inconsistency or Critical severity	Results shown
Pending decision (TBC)	Medium-only or combined-flag case (Q-01, Q-02)	Results shown
Presented	Findings and suggestion displayed	Seller decision / no decision
Decided	Accept, Edit or Ignore recorded	Session ends
Closed	Session-local end state	Final state

Design notes and traceability

Requirements	FR-014, FR-015, FR-019, FR-020, FR-023 to FR-025, NFR-REL-01, NFR-REL-04
Objectives	OBJ-06, OBJ-08
Open items	Q-01 and Q-02 keep one state orange until decided.

9.4 Sequence Diagrams

Six sequence diagrams cover the main analysis and five alternative or exceptional scenarios. Messages are numbered; dashed lines are returns.

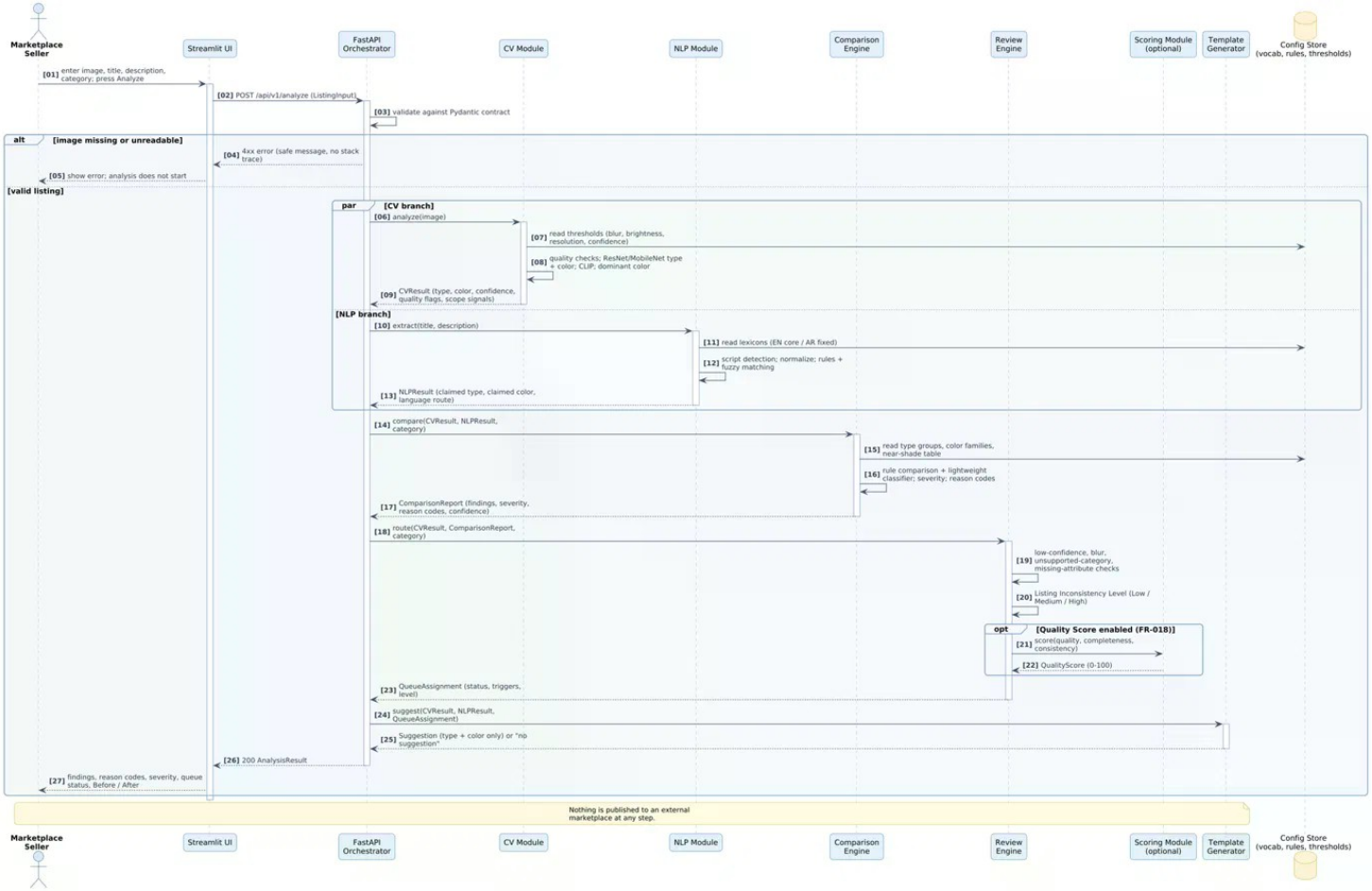


Figure 14. *Sequence diagram D11a: main analysis (UC-01 to UC-09)*

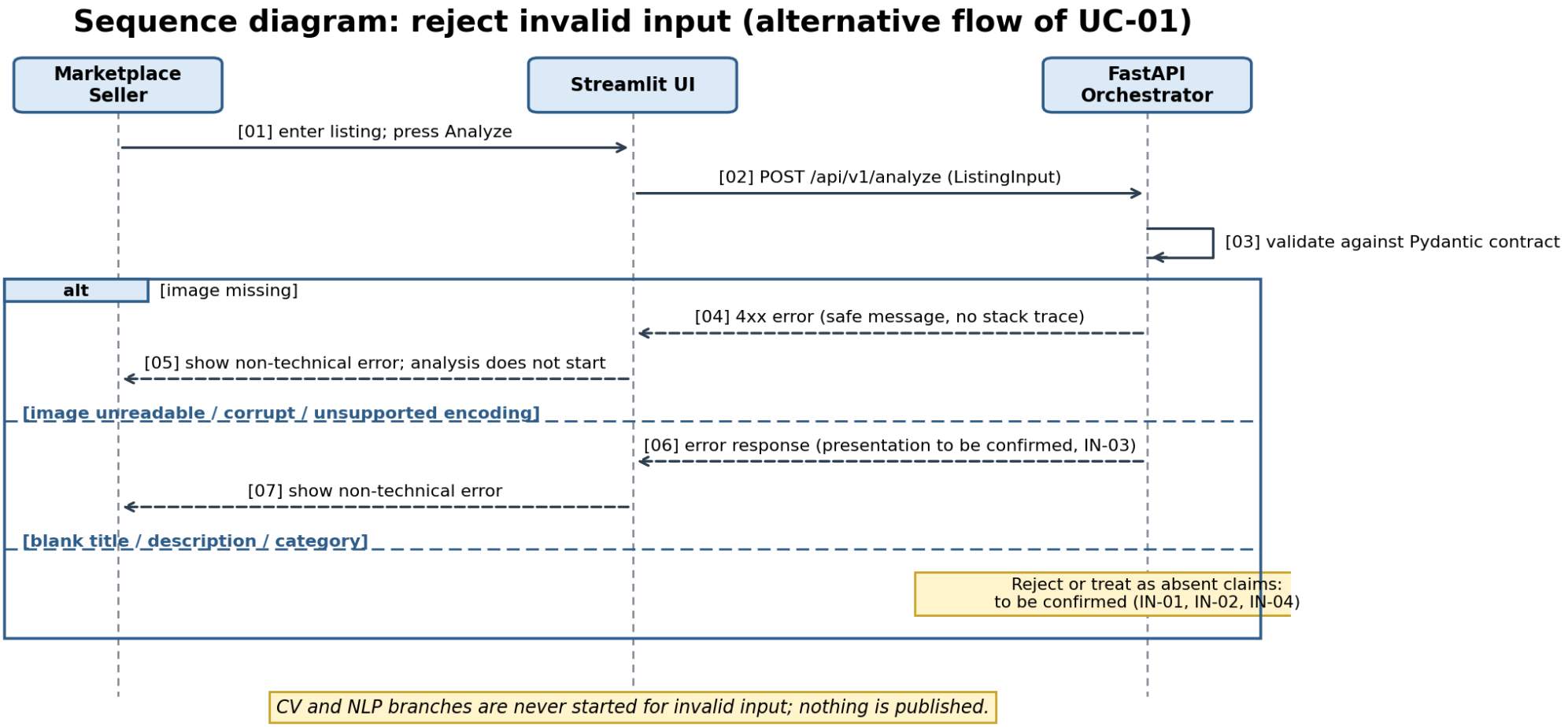


Figure 15. *Sequence diagram D11b: reject invalid input*

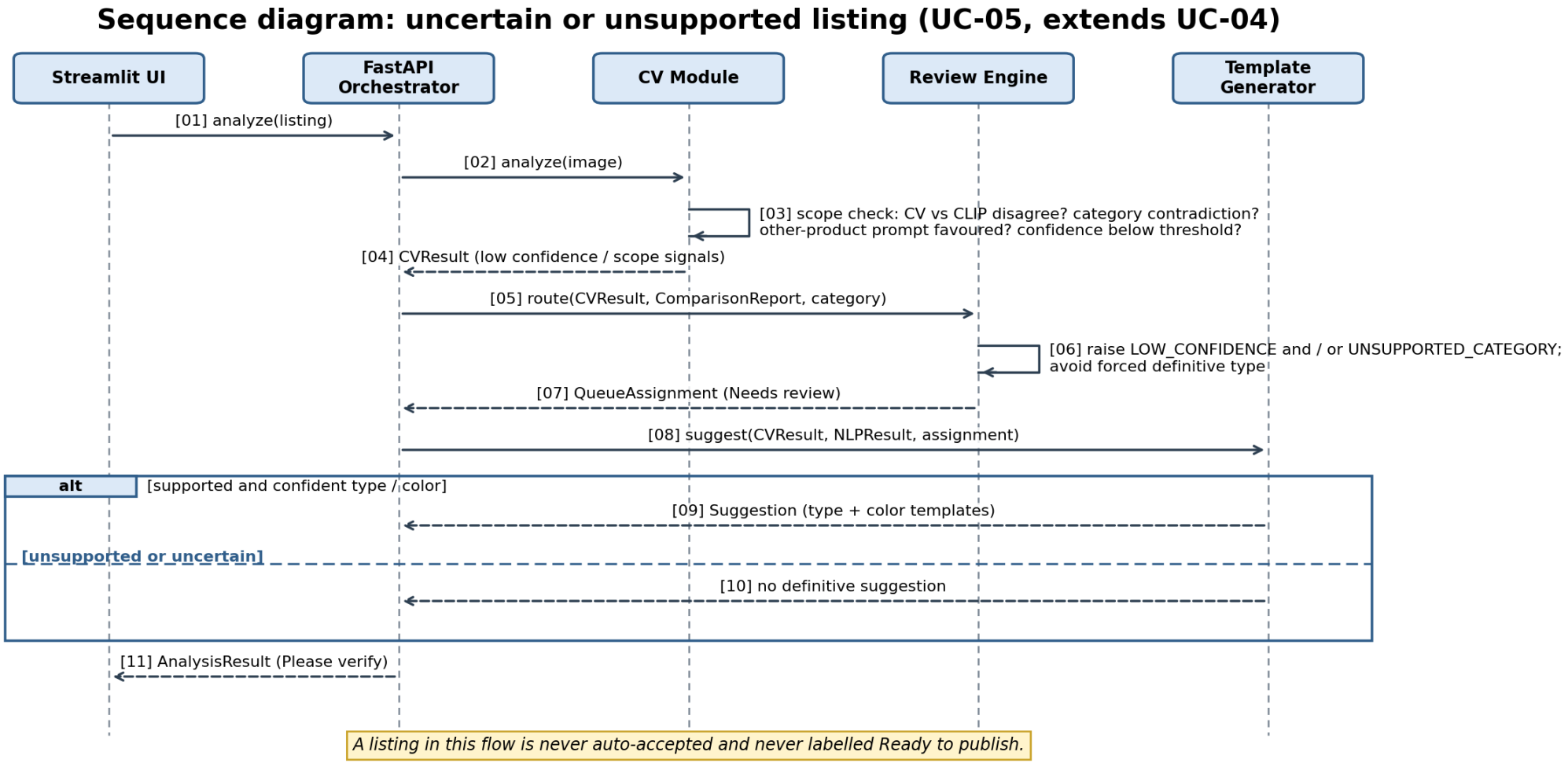


Figure 16. *Sequence diagram D11c: uncertain or unsupported listing*

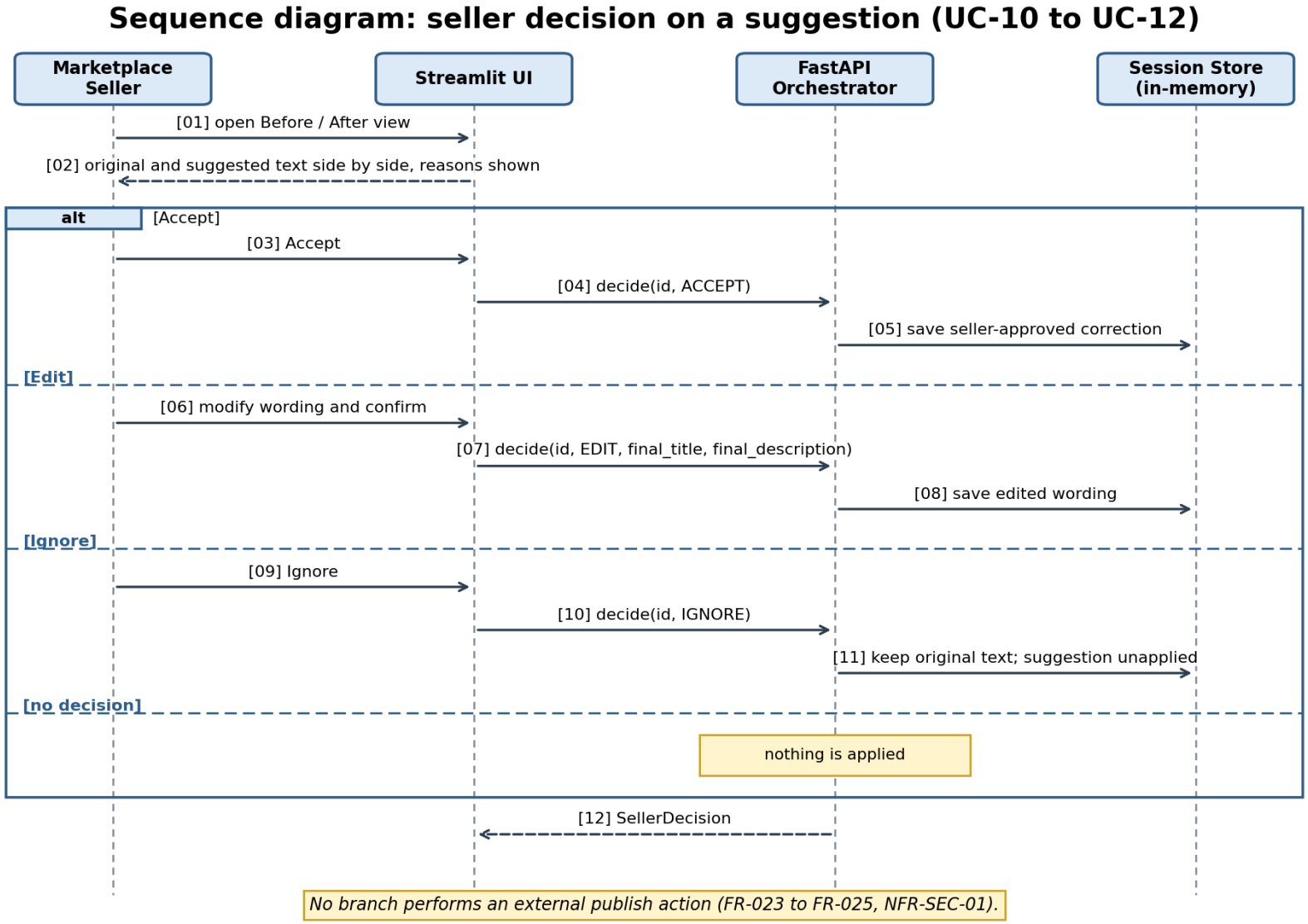


Figure 17. *Sequence diagram D11d: seller decision*

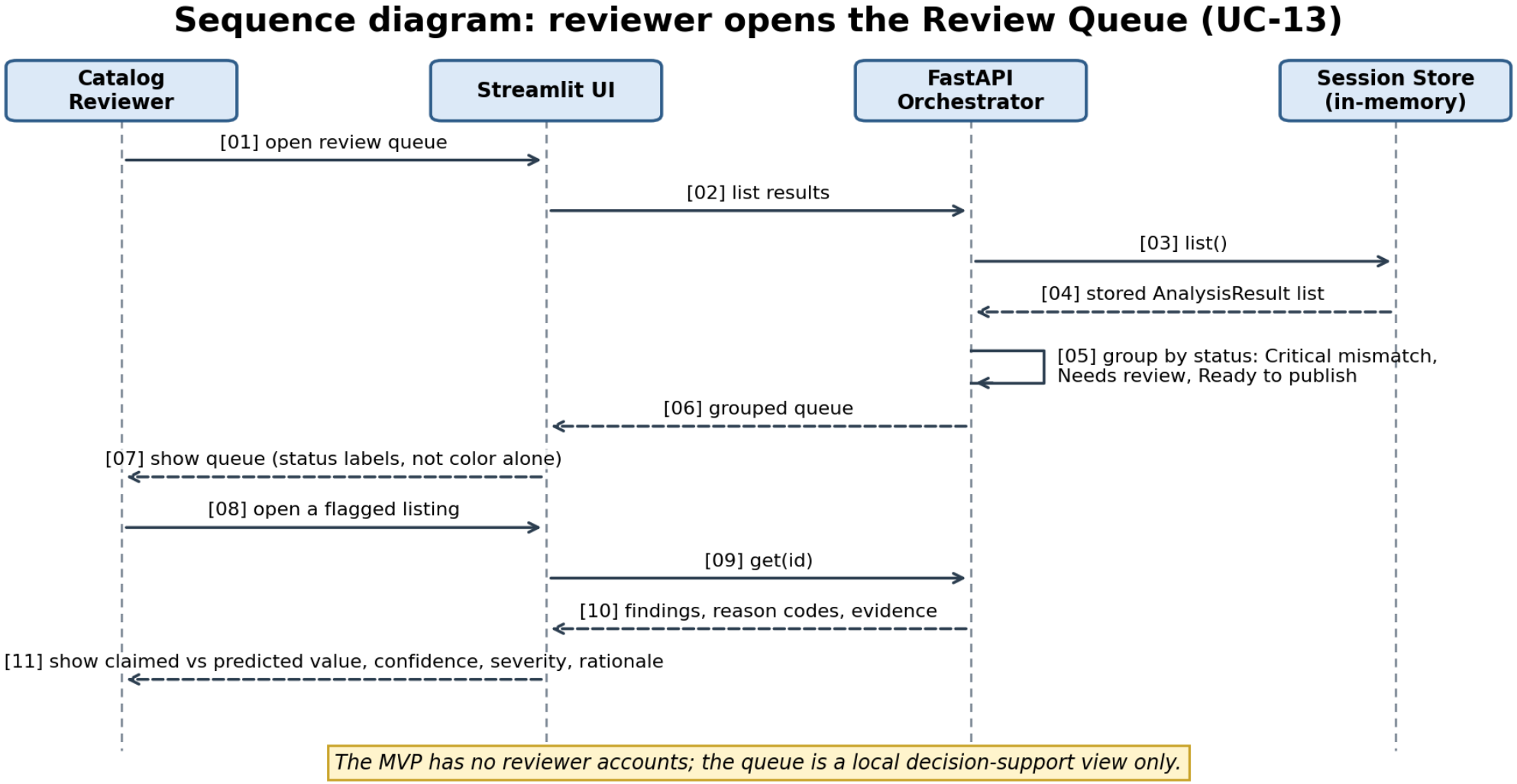


Figure 18. *Sequence diagram D11e: reviewer opens the Review Queue*

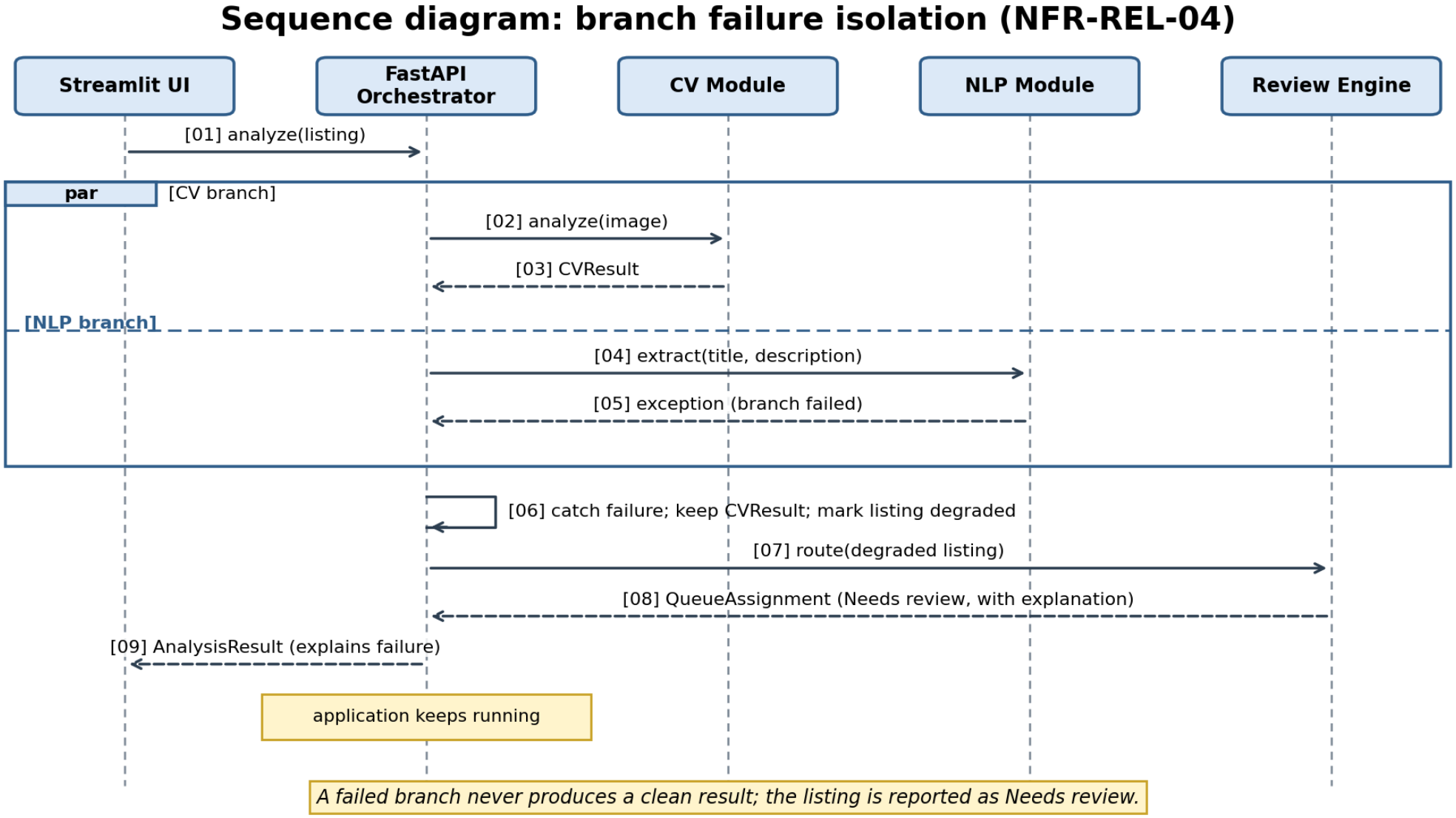


Figure 19. *Sequence diagram D11f: branch failure isolation*

Explanation of the sequence diagrams

Diagram	Scenario	Key messages	Outcome	Requirements
D11a	Main analysis	POST /api/v1/analyze; validate; parallel analyze(image) and extract(title, description); compare; route; optional score; suggest; AnalysisResult	Findings, reason codes, severity, queue status, Before / After. Nothing is published (note on diagram).	FR-001 to FR-022, UC-01 to UC-09
D11b	Invalid input	Validate; 4xx safe message; show non-technical error	Analysis does not start; unreadable file, empty fields and malformed category are open (IN-01 to IN-04)	FR-001, NFR-SEC-02 to SEC-04
D11c	Uncertain or unsupported	Scope check in CV; route raises LOW_CONFIDENCE or UNSUPPORTED_CATEGORY; suggest returns a suggestion or none	Needs review with "Please verify"; no forced type; never Ready to publish	FR-014, FR-015, UC-05, NFR-REL-01
D11d	Seller decision	decide(id, ACCEPT / EDIT / IGNORE); save decision	Seller-approved, edited or unapplied text; no decision applies nothing; no publish call	FR-023 to FR-025, UC-12, NFR-SEC-01
D11e	Reviewer queue	list(); group by status; get(id)	Grouped queue and evidence per listing; no reviewer accounts in the MVP	FR-016, FR-017, FR-019, FR-020, UC-13
D11f	Branch failure	Exception from one branch; catch; route degraded listing	Needs review with explanation; application keeps running	NFR-REL-04, TAC-11

9.5 Communication Diagram

The communication diagram shows the same main analysis as D11a but emphasises which objects collaborate, with sequence carried by message numbers.

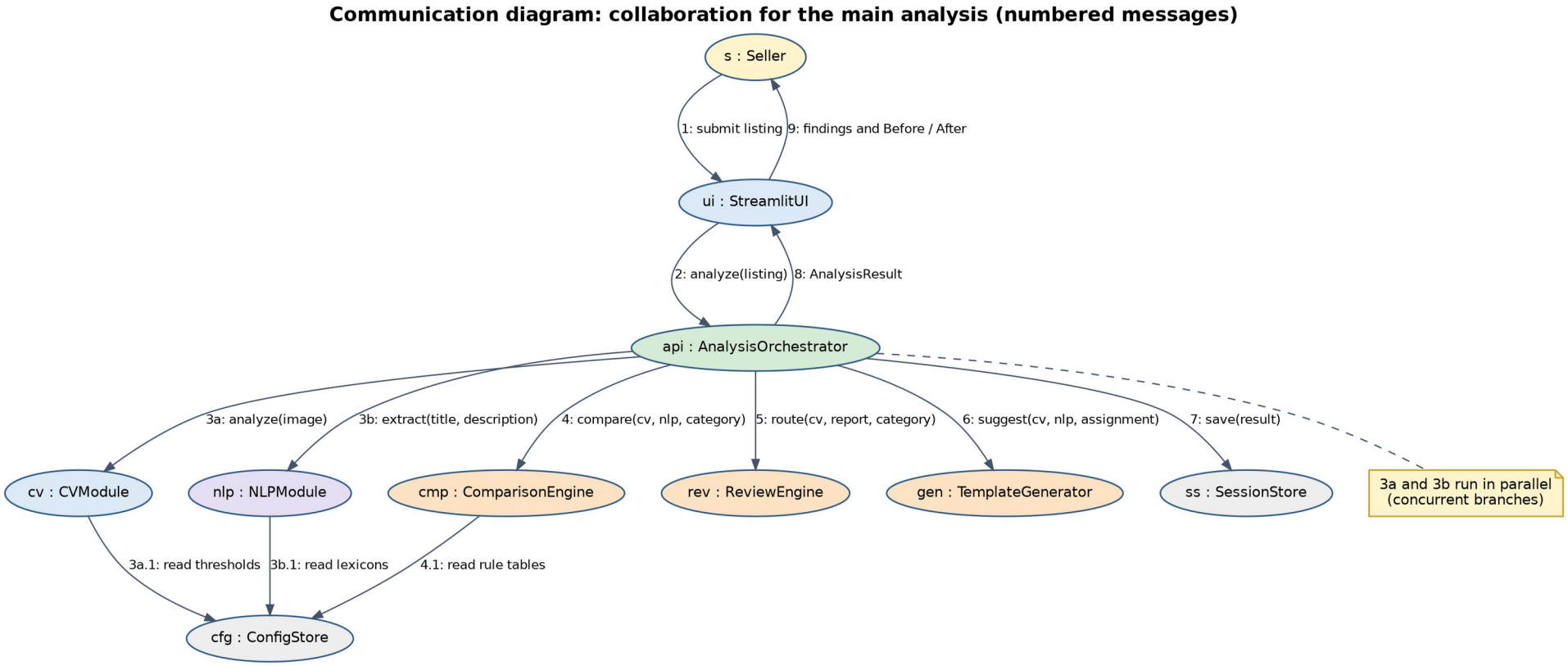


Figure 20. *Communication diagram: collaboration for the main analysis (numbered messages)*

How to read the diagram

- Messages 1 to 9 run from the seller through the UI to the orchestrator and back; 3a and 3b are concurrent branches.
- The orchestrator calls the CV module (3a), the NLP module (3b), the comparison engine (4), the review engine (5), the template generator (6) and the session store (7).
- Modules read configuration only: thresholds (3a.1), lexicons (3b.1) and rule tables (4.1).

Design notes and traceability

Requirements	FR-001 to FR-022, NFR-PERF-02
Objectives	OBJ-01

9.6 Interaction Overview Diagram

The interaction overview diagram is an activity diagram whose nodes are interactions. It ties the sequence diagrams together into the overall behavior of a listing.

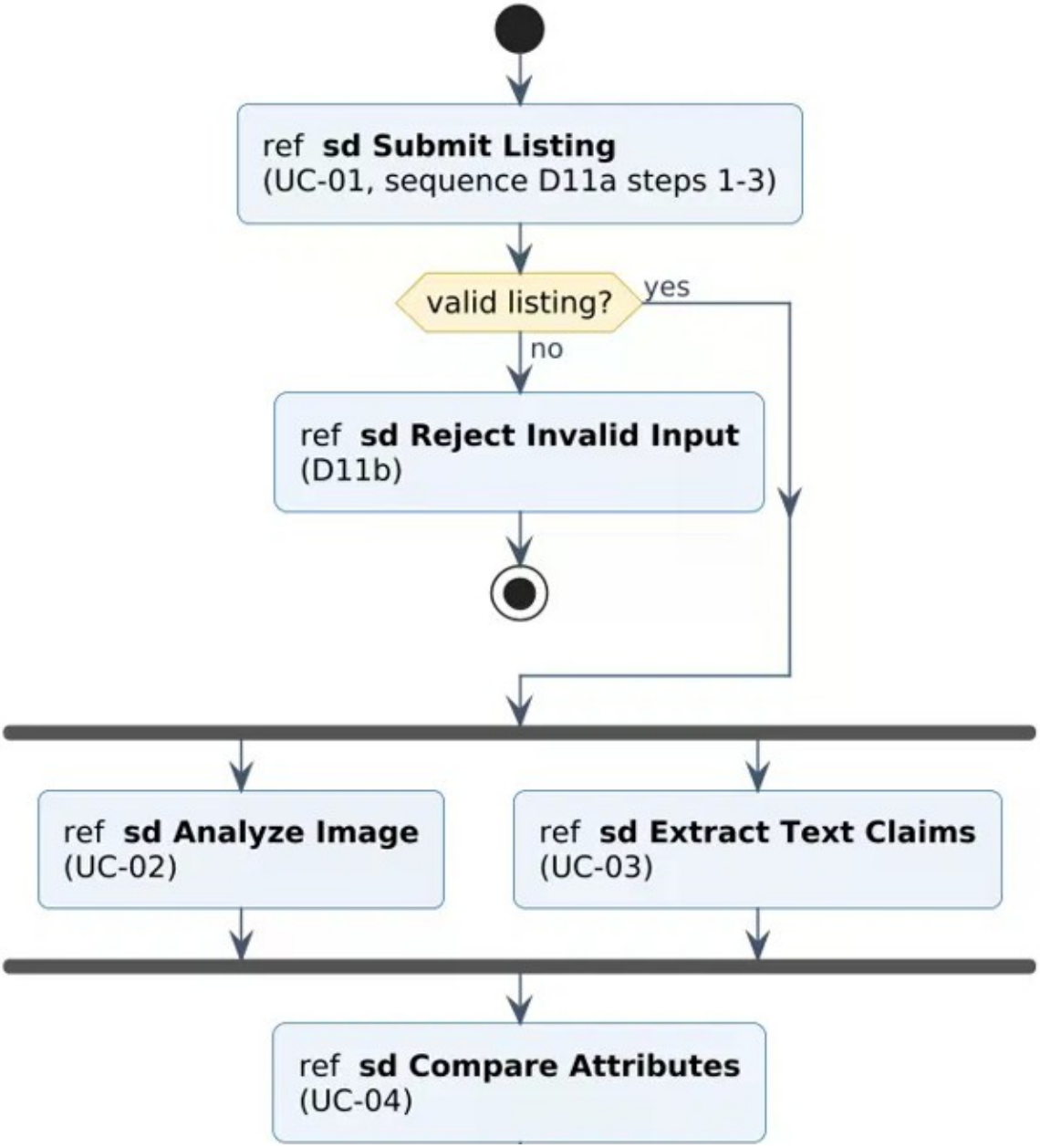


Figure 21. *Interaction overview diagram (part 1 of 2)*

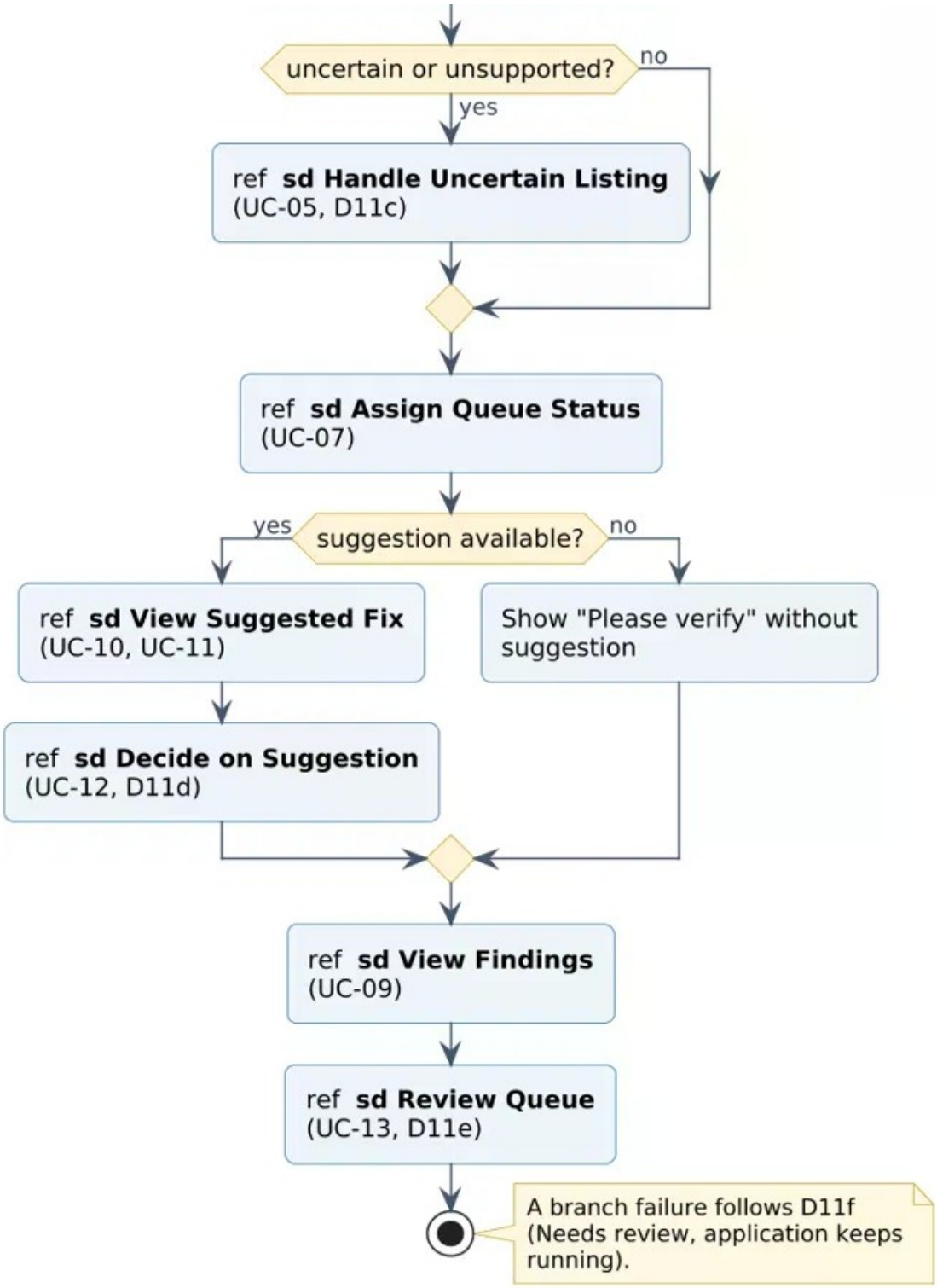


Figure 22. *Interaction overview diagram (part 2 of 2)*

How to read the diagram

- Submit Listing (D11a steps 1 to 3) is followed by the decision "valid listing?"; the no branch references Reject Invalid Input (D11b) and stops.
- A fork runs Analyze Image (UC-02) and Extract Text Claims (UC-03) in parallel; Compare Attributes (UC-04) follows, then Handle Uncertain Listing (UC-05, D11c) when uncertain or unsupported.
- Assign Queue Status (UC-07) leads to either View Suggested Fix and Decide on Suggestion (UC-10 to UC-12, D11d) or "Please verify" without a suggestion.
- View Findings (UC-09) and Review Queue (UC-13, D11e) close the flow; a branch failure follows D11f.

Design notes and traceability

Requirements	FR-001 to FR-025, NFR-REL-04
Objectives	OBJ-01, OBJ-06, OBJ-08

9.7 Timing Diagram

The timing diagram shows the ordering and overlap of state changes across the UI, orchestrator, branches, comparison and template components during one analysis.

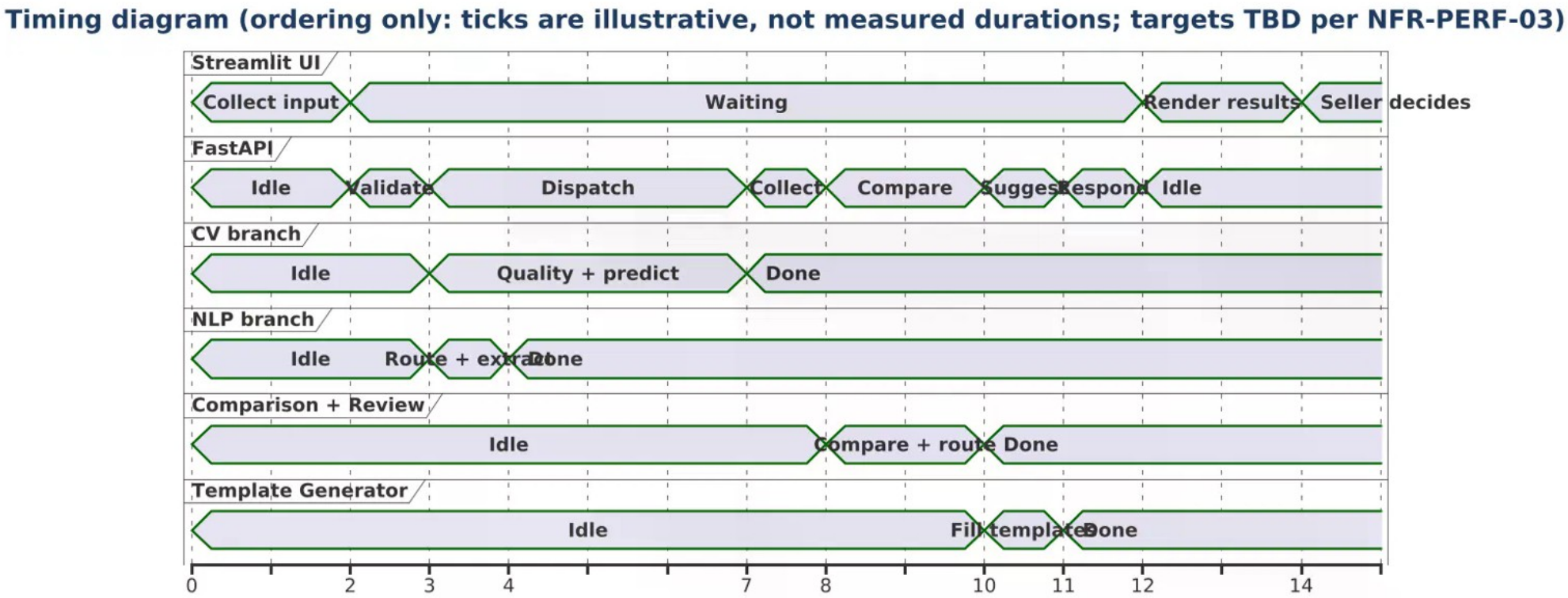


Figure 23. *Timing diagram (ordering only: ticks are illustrative, not measured durations)*

How to read the diagram

- Time runs left to right in illustrative ticks. The UI waits while FastAPI validates, dispatches and joins.
- The CV branch (quality, predict) and NLP branch (extract) overlap, and the NLP branch finishes earlier; total latency is bounded by the slower branch plus comparison (NFR-PERF-02).
- Comparison and review run after the join, then the template generator, then the UI renders results and the seller decides.

Design notes and traceability

Requirements	NFR-PERF-01 to NFR-PERF-03, TAC-01
Objectives	OBJ-01, OBJ-09
Caution	Targets are TBD per NFR-PERF-03; the diagram claims no duration.

10. Other System Analysis and Design Diagrams

These diagrams analyse the system from the data, process, decision, module and schedule points of view. They complement the UML models and are kept consistent with them (section 11.2).

10.1 Context Diagram

The context diagram (DFD level 0) shows the whole system as one process and every data flow with the outside world.

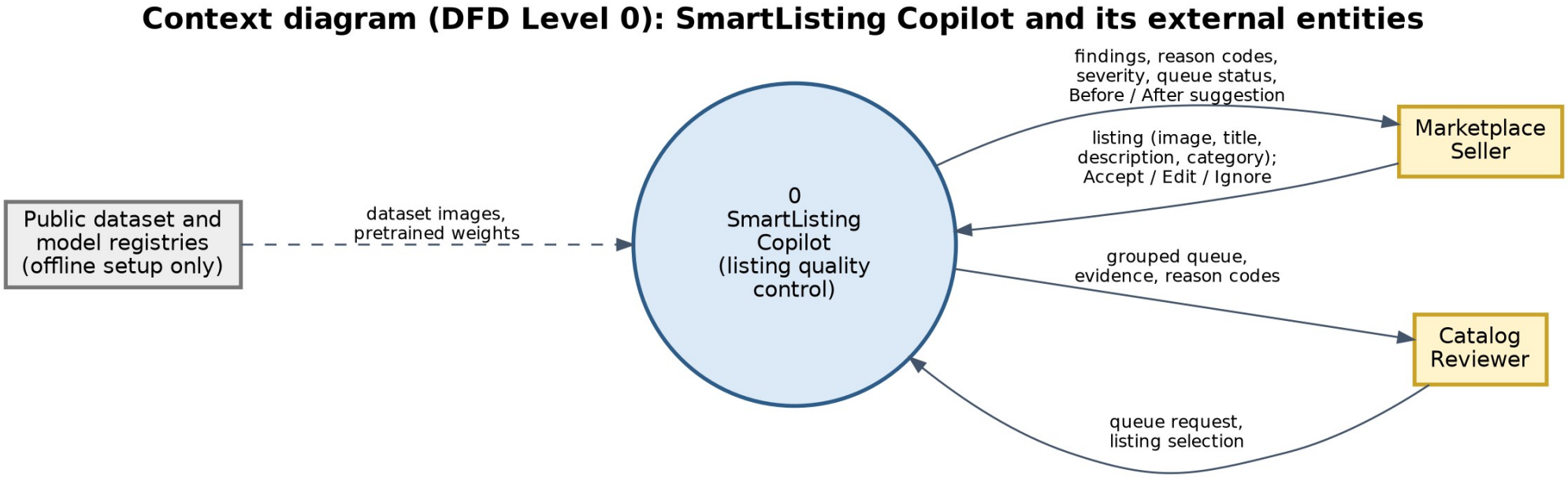


Figure 24. *Context diagram (DFD level 0)*

How to read the diagram

- External entities: Marketplace Seller, Catalog Reviewer and, for offline setup only, public dataset and model registries.
- The seller sends the listing and the Accept / Edit / Ignore decision and receives findings, reason codes, severity, queue status and the Before / After suggestion.
- The reviewer sends a queue request and receives the grouped queue with evidence. No flow leads to an external marketplace.

Design notes and traceability

Requirements	FR-001 to FR-003, FR-016 to FR-025, NFR-SEC-06, NFR-SEC-07
Objectives	OBJ-01

10.2 Data Flow Diagrams (DFD)

The level 1 DFD decomposes the system into eight processes, three data stores and two external entities. Level 2 diagrams then explode the two most decision-heavy processes, 4.0 and 5.0.

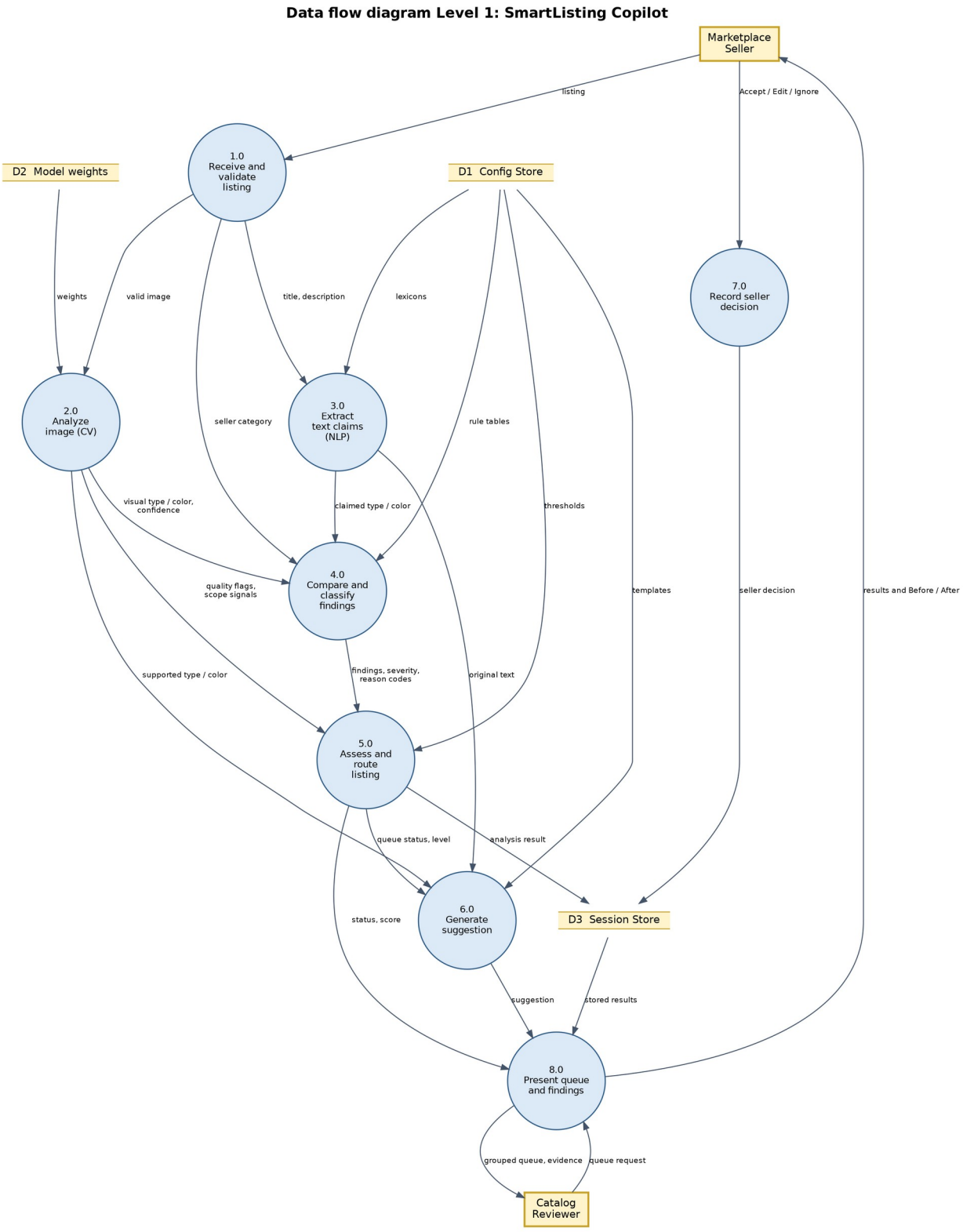


Figure 25. *Data flow diagram, level 1*

How to read the diagram

- Processes: 1.0 Receive and validate listing; 2.0 Analyze image (CV); 3.0 Extract text claims (NLP); 4.0 Compare and classify findings; 5.0 Assess and route listing; 6.0 Generate suggestion; 7.0 Record seller decision; 8.0 Present queue and findings.
- Stores: D1 Config Store (vocabularies, rule tables, thresholds, templates), D2 Model weights, D3 Session Store (results, decisions).

- Balancing: the inputs and outputs of the context diagram (listing, decision, results, queue request, grouped queue) reappear unchanged at level 1.

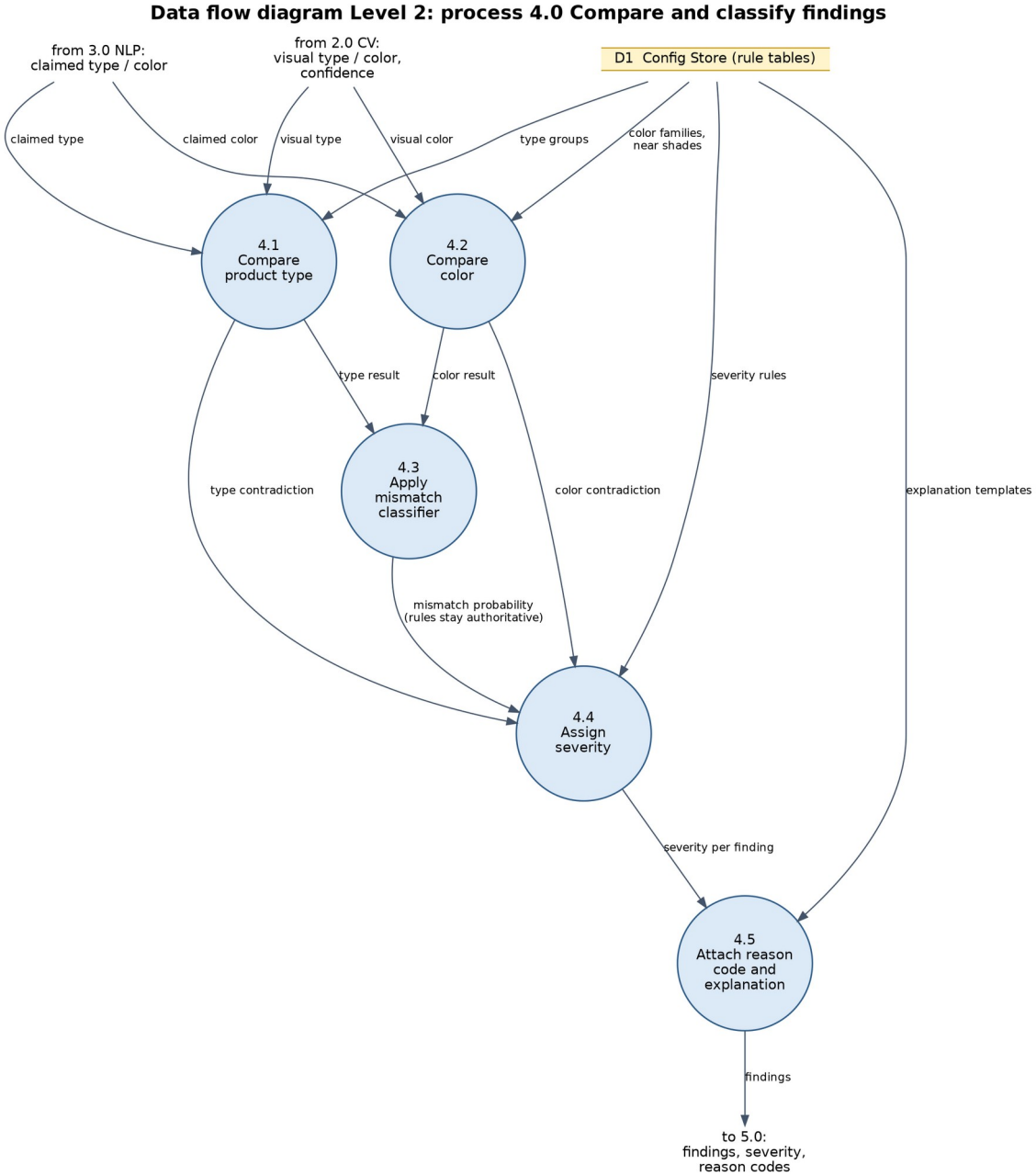


Figure 26. *Data flow diagram, level 2: process 4.0 Compare and classify findings*

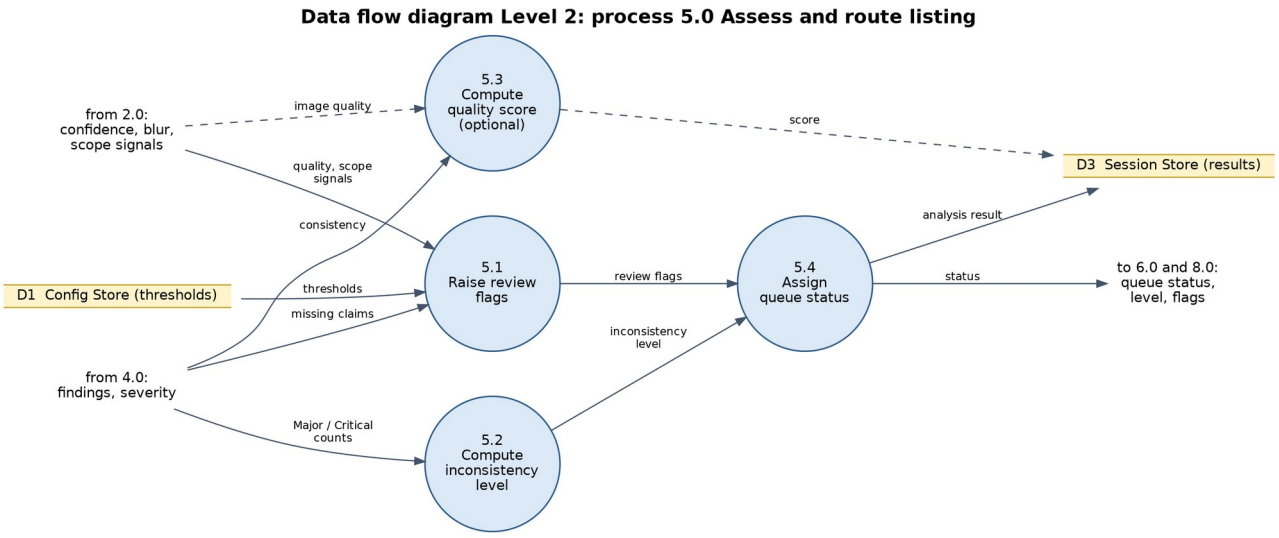


Figure 27. *Data flow diagram, level 2: process 5.0 Assess and route listing*

Data dictionary of the main flows

Data flow	Content	Source to destination
listing	image, title, description, category	Seller to 1.0
valid image / title, description / seller category	Validated parts of the listing	1.0 to 2.0, 3.0, 4.0
visual type / color, confidence	Predictions from CV with confidence	2.0 to 4.0
quality flags, scope signals	Blur, resolution, brightness, CV vs CLIP, category contradiction, other-product preference	2.0 to 5.0
claimed type / color	Normalised text claims and language route	3.0 to 4.0
findings, severity, reason codes	Per-finding attribute, claimed and predicted values, confidence, severity, code, explanation	4.0 to 5.0
queue status, level, flags	Ready / Needs review / Critical mismatch, Low / Medium / High, review flags	5.0 to 6.0, 8.0
suggestion	Suggested title and description, change items	6.0 to 8.0
seller decision	ACCEPT, EDIT or IGNORE with final text	Seller to 7.0 to D3
grouped queue, evidence	Listings by status with reason codes	8.0 to Reviewer

Design notes and traceability

Requirements	FR-001 to FR-025
Objectives	OBJ-01 to OBJ-08
Status	Level 1 redrawn for readability; level 2 diagrams are new. Process 4.0 shows the classifier flowing into severity because rules remain authoritative.

10.3 Entity-Relationship Diagram (ERD)

The ERD gives the logical data model of one analysis session. The MVP keeps data in memory; the model shows how the same data would be stored and how entities relate.

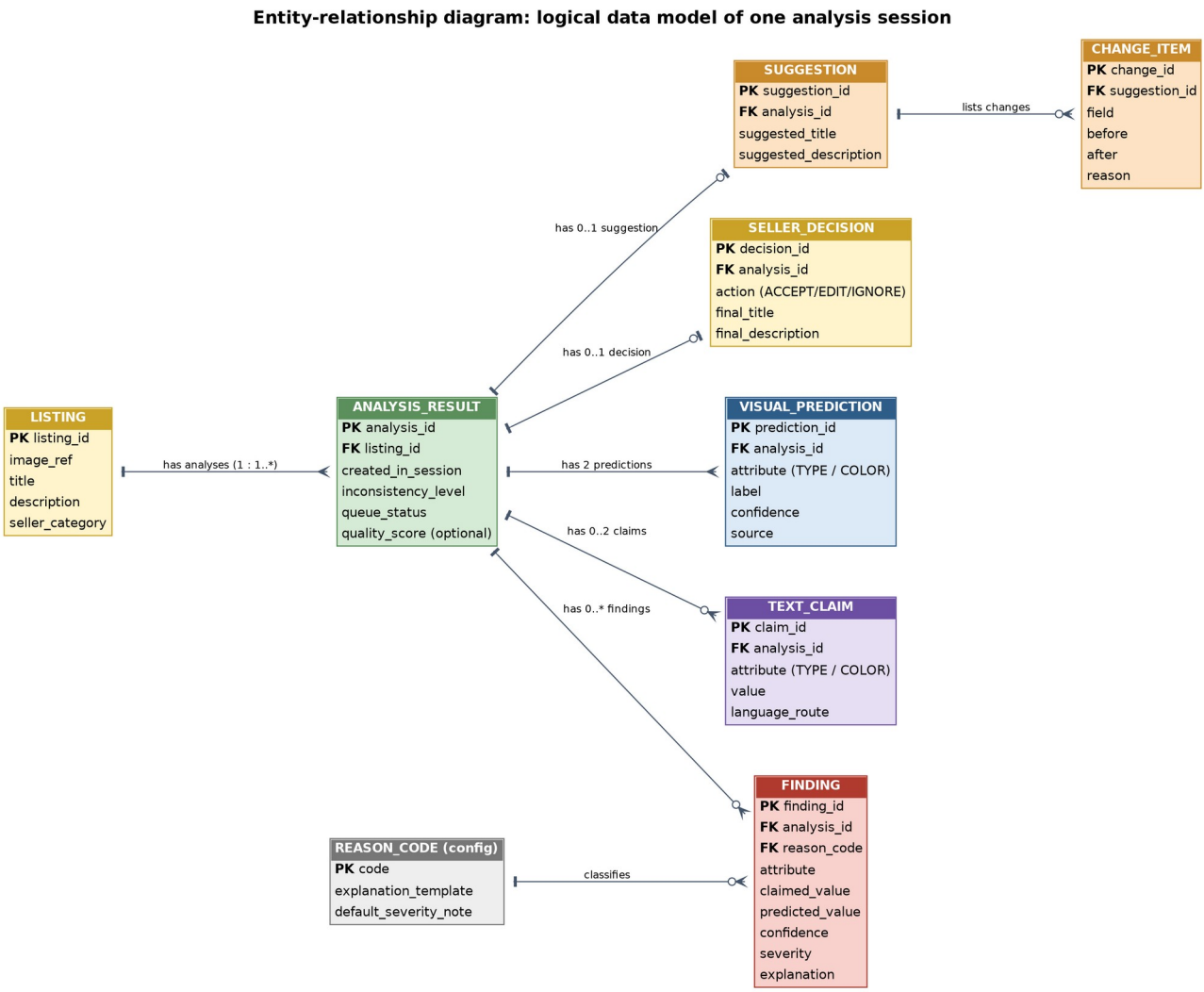


Figure 28. *Entity-relationship diagram (crow-foot notation)*

How to read the diagram

- LISTING 1 to 1..* ANALYSIS_RESULT; an analysis has two VISUAL_PREDICTION rows (type and color), zero to two TEXT_CLAIM rows and zero or more FINDING rows.
- FINDING references the configuration lookup REASON_CODE; an analysis has at most one SUGGESTION (with zero or more CHANGE_ITEM rows) and at most one SELLER_DECISION.
- REASON_CODE, type groups and color families are configuration files, not session data.

Entity dictionary

Entity	Key attributes	Notes
LISTING	listing_id, image_ref, title, description, seller_category	One submitted listing
ANALYSIS_RESULT	analysis_id, listing_id, inconsistency_level, queue_status, quality_score	Aggregate root; score optional; created_in_session
VISUAL_PREDICTION	attribute, label, confidence, source	source RESNET_MOBILENET or CLIP_ZERO_SHOT
TEXT_CLAIM	attribute, value, language_route	Only type and color
FINDING	reason_code, claimed_value, predicted_value, confidence, severity, explanation	One per surfaced warning
REASON_CODE	code, explanation_template	Six fixed codes
SUGGESTION / CHANGE_ITEM	suggested_title, suggested_description / field, before, after, reason	Template output

Entity	Key attributes	Notes
SELLER_DECISION	action, final_title, final_description	No publish attribute

Design notes and traceability

Requirements	FR-009 to FR-025, NFR-REL-08, CON-03
Objectives	OBJ-04, OBJ-05, OBJ-07, OBJ-08
Open items	Duration and representation of seller decisions (FRS 12.4) is undecided; queue_status has no value for Medium-only (Q-01).

10.4 Flowcharts

Program flowcharts show the decision logic step by step. Flowchart A covers validation, parallel analysis and comparison; flowchart B covers review routing, suggestion and the seller decision.

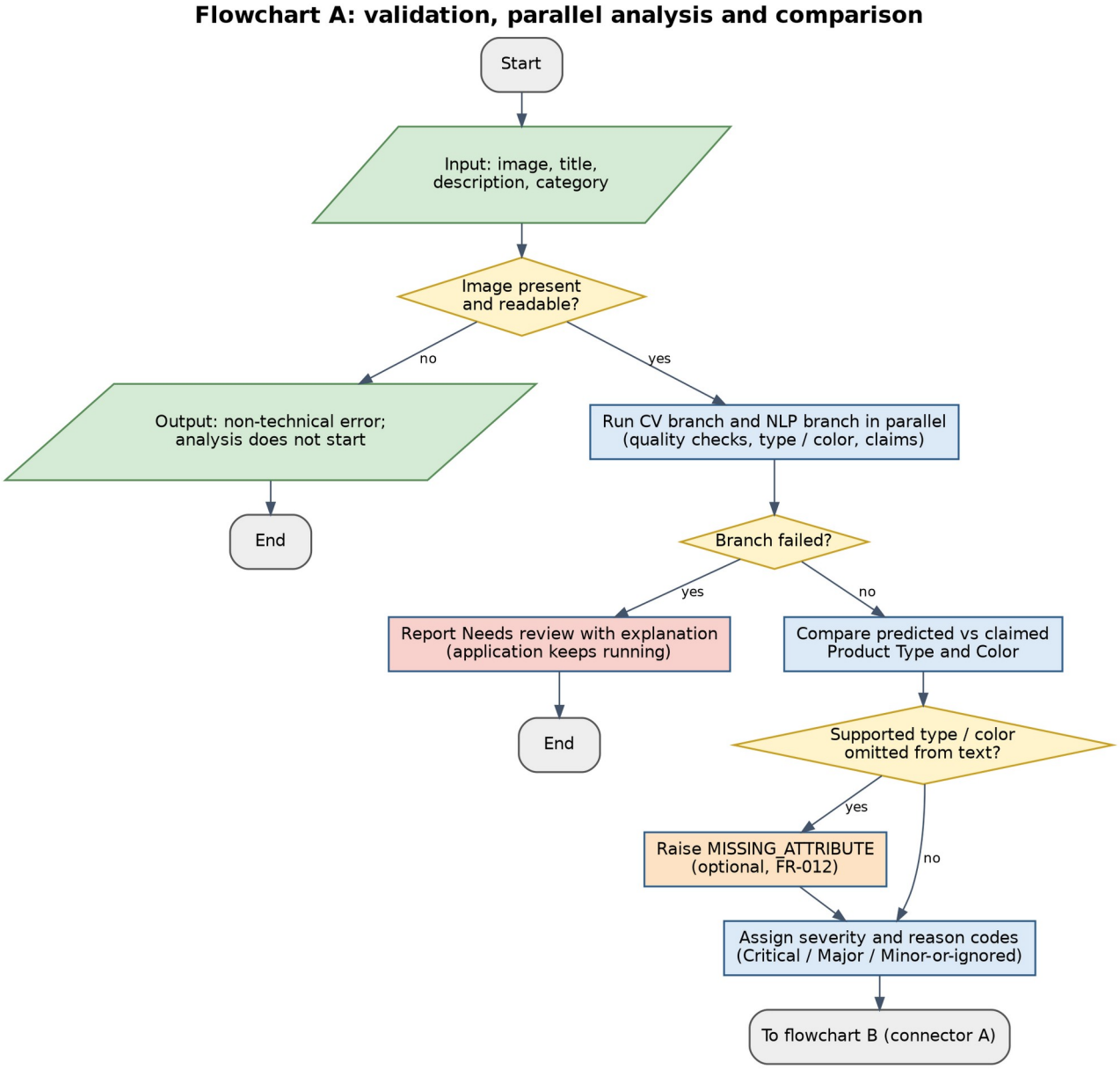


Figure 29. *Flowchart A: validation, parallel analysis and comparison*

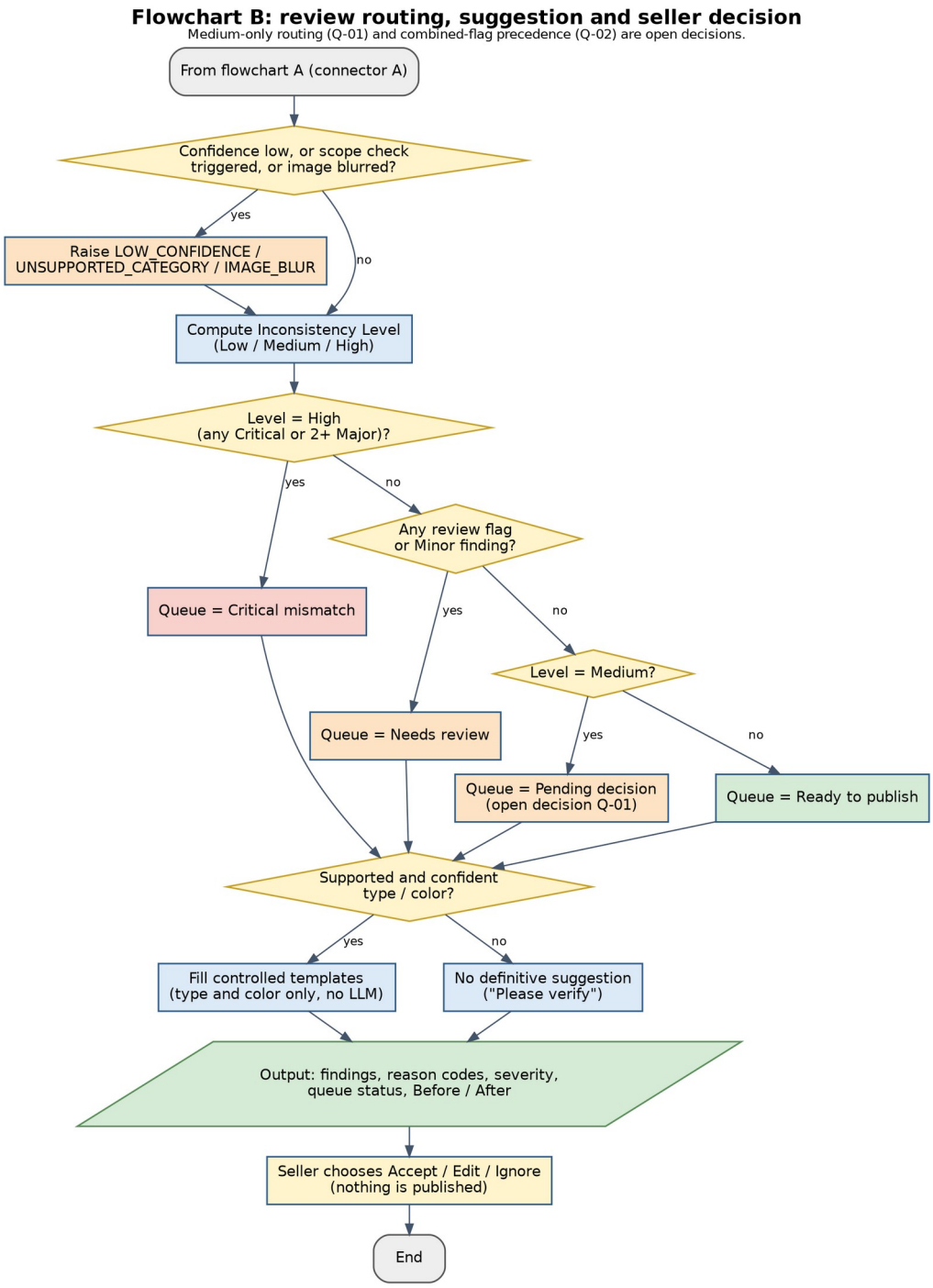


Figure 30. *Flowchart B: review routing, suggestion and seller decision*

How to read the diagram

- Terminators are rounded boxes, processes rectangles, decisions diamonds and input / output parallelograms.
- Flowchart A ends at connector A; flowchart B starts from it. A failed branch ends in Needs review with an explanation and the application keeps running.
- Orange boxes are flags or open decisions: Medium-only routing (Q-01) and combined-flag precedence (Q-02).

Design notes and traceability

Requirements	FR-001, FR-009 to FR-021, NFR-REL-01, NFR-REL-04
Objectives	OBJ-04, OBJ-06, OBJ-07

10.5 System Flowchart

The system flowchart shows how processes, input and output devices and stored data cooperate in the local prototype.

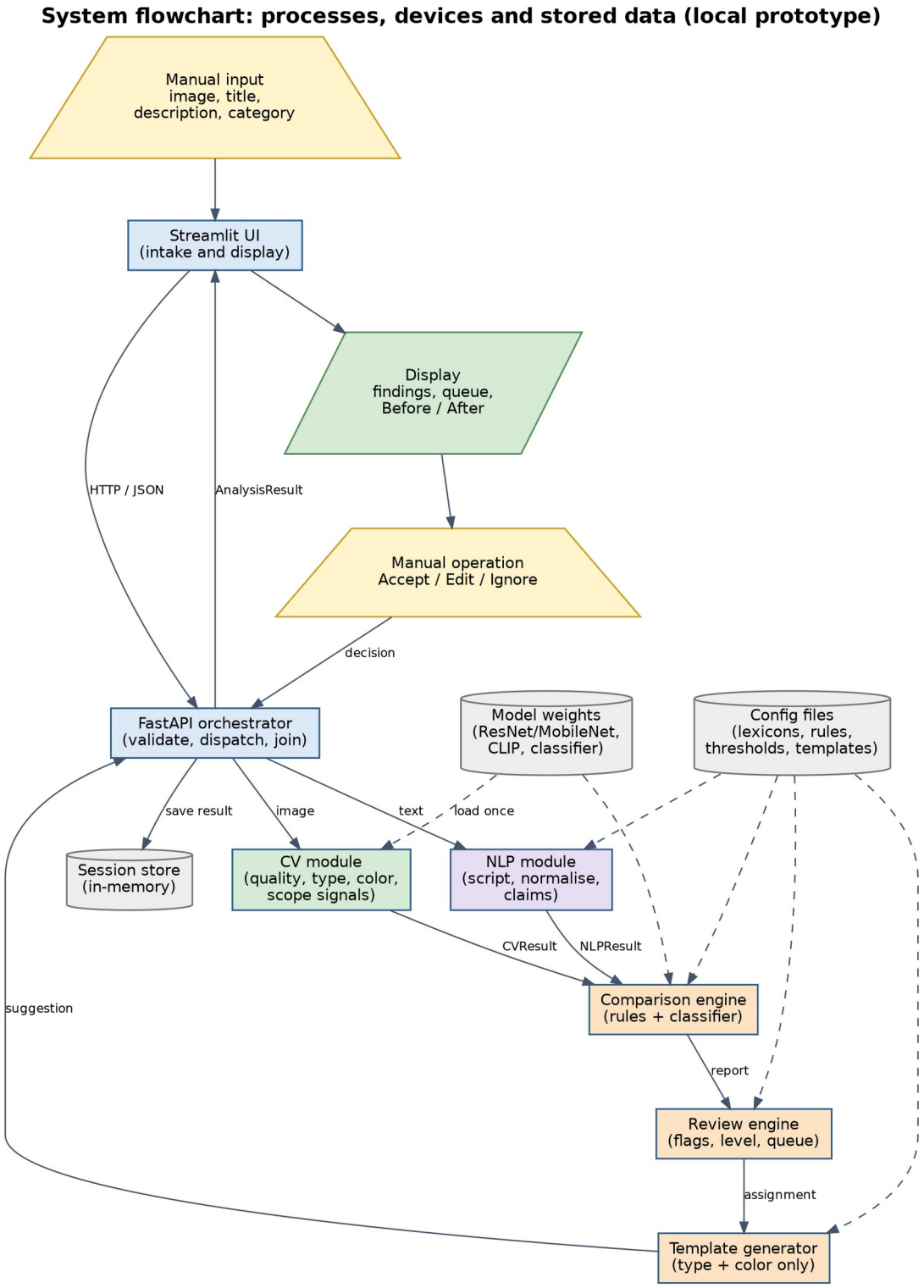


Figure 31. *System flowchart: processes, devices and stored data*

How to read the diagram

- Manual operations (trapezoids) are the seller entering the listing and choosing Accept / Edit / Ignore.
- Cylinders are stores: model weights loaded once, configuration files read at run time and the in-memory session store.
- The UI calls FastAPI, which fans out to CV and NLP, then through comparison, review and template generation, and returns an AnalysisResult for display.

Design notes and traceability

Requirements	NFR-PERF-08, NFR-SEC-06, NFR-TEC-02, NFR-TEC-03
Objectives	OBJ-01

10.6 Document Flowchart

The document flowchart follows the documents and records that pass between the seller, the system and the reviewer.

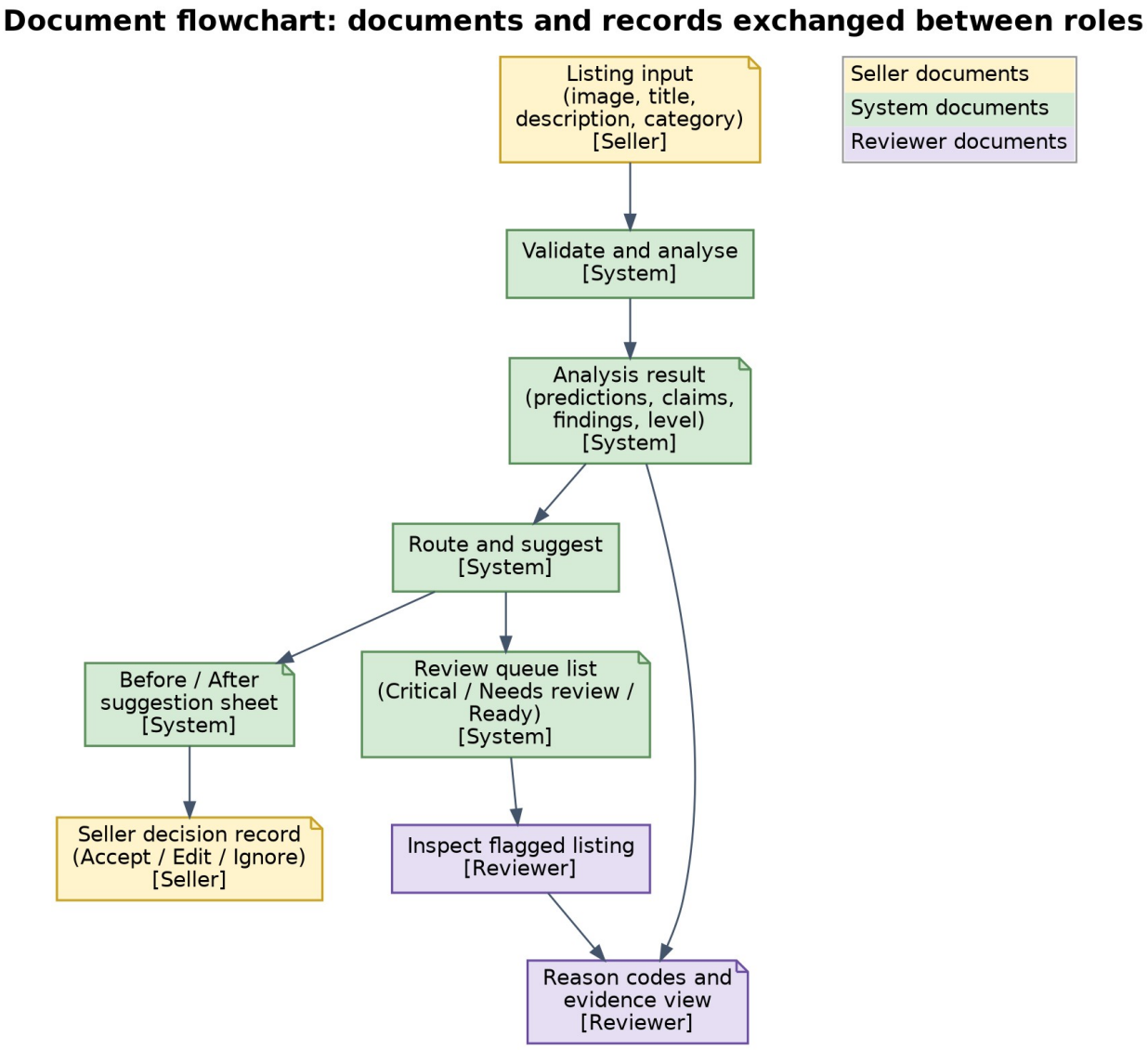


Figure 32. *Document flowchart*

How to read the diagram

- The seller supplies a listing input document; the system produces an analysis result, a Before / After suggestion sheet and a review queue list.
- The seller returns a decision record; the reviewer inspects flagged listings and reads the reason codes and evidence view.
- Colors show who owns each document: seller (yellow), system (green), reviewer (purple). No document goes to an external marketplace.

Design notes and traceability

Requirements	FR-016 to FR-025

Objectives	OBJ-05, OBJ-06, OBJ-08

10.7 Decision Tree

The decision tree shows how the Inconsistency Level and the review flags lead to the Review Queue status.

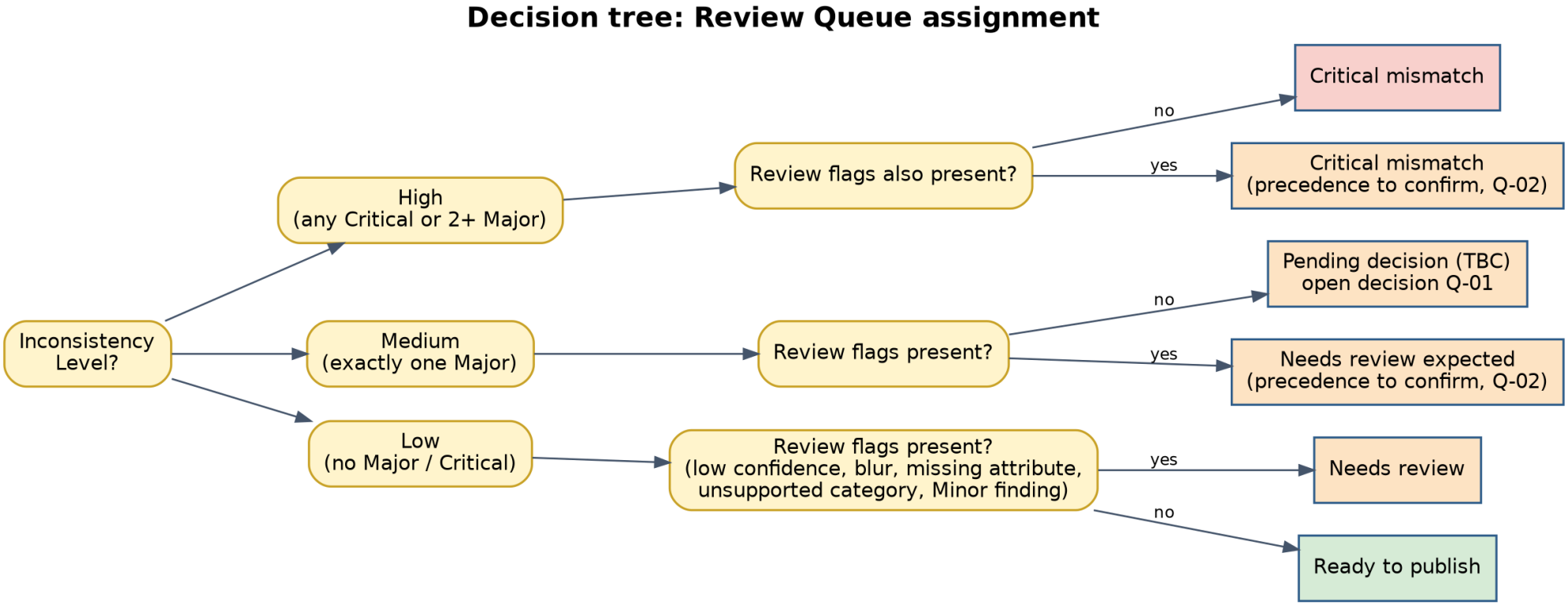


Figure 33. *Decision tree: Review Queue assignment*

How to read the diagram

- Level High (any Critical or two or more Major) leads to Critical mismatch; when review flags also exist the precedence is still to be confirmed (Q-02).
- Level Medium (exactly one Major) with no flags is "Pending decision" (Q-01); with a flag it is expected to be Needs review but precedence needs confirmation (Q-02).
- Level Low with a review flag (low confidence, blur, missing attribute, unsupported category or a Minor finding) is Needs review; Low with no flag is Ready to publish.

Design notes and traceability

Requirements	FR-019, FR-020, AC-020, TAC-UC07
Objectives	OBJ-06
Rule	Specified results are in green or red; orange marks behavior the sources leave open. No default is assumed.

10.8 Decision Tables

Decision tables list every combination of conditions with its action, which makes gaps visible. Four tables cover severity, inconsistency level, queue assignment and suggestion generation. Orange cells are open decisions.

Table DT-1: Mismatch severity (FR-010, FR-011, FR-017)

Rule	Attribute	Condition	Severity	Reason code
1	Product Type	Contradiction between different product groups	Critical	TYPE_MISMATCH
2	Product Type	Contradiction within the same group	Major	TYPE_MISMATCH
3	Color	Different color families	Major	COLOR_MISMATCH
4	Color	Near shade	Minor or ignored	COLOR_MISMATCH (if kept as a finding)
5	Type or Color	Supported value absent from text	No mismatch; review flag	MISSING_ATTRIBUTE (optional)

Table DT-2: Listing Inconsistency Level (FR-019)

Rule	Critical findings	Major findings	Level
1	0	0	Low
2	0	1	Medium
3	0	2 or more	High
4	1 or more	any	High

Table DT-3: Review Queue assignment (FR-020)

Rule	Inconsistency level	Review flag present	Queue status	Status of the rule
1	Low	No	Ready to publish	Specified
2	Low	Yes	Needs review	Specified
3	Medium (one Major)	No	Pending decision (TBC)	Open Q-01
4	Medium	Yes	To be confirmed	Open Q-02
5	High (Critical or 2+ Major)	No	Critical mismatch	Specified
6	High	Yes	To be confirmed	Open Q-02

Table DT-4: Suggestion generation (FR-021, FR-014, FR-015)

Rule	Type supported and confident	Color supported and confident	Result
1	Yes	Yes	Template with type and color
2	Yes	No	Template with type only
3	No	Yes	Template with color only
4	No	No	No definitive suggestion ("Please verify")

Each table is exhaustive for its conditions. Where a rule is shaded orange, the sources deliberately leave behavior undecided, so tests for those rules stay blocked until the decision is recorded in the change log. DT-4 rules 2 and 3 follow from the type-and-color-only template rule; the source states the principle but not each combination, so they should be confirmed with the NLP lead.

Requirements	FR-010, FR-011, FR-014, FR-015, FR-017, FR-019, FR-020, FR-021
Objectives	OBJ-04, OBJ-05, OBJ-06, OBJ-07

10.9 Structure Chart

The structure chart shows the module hierarchy of the analysis program, which module calls which, and the data passed down and returned.

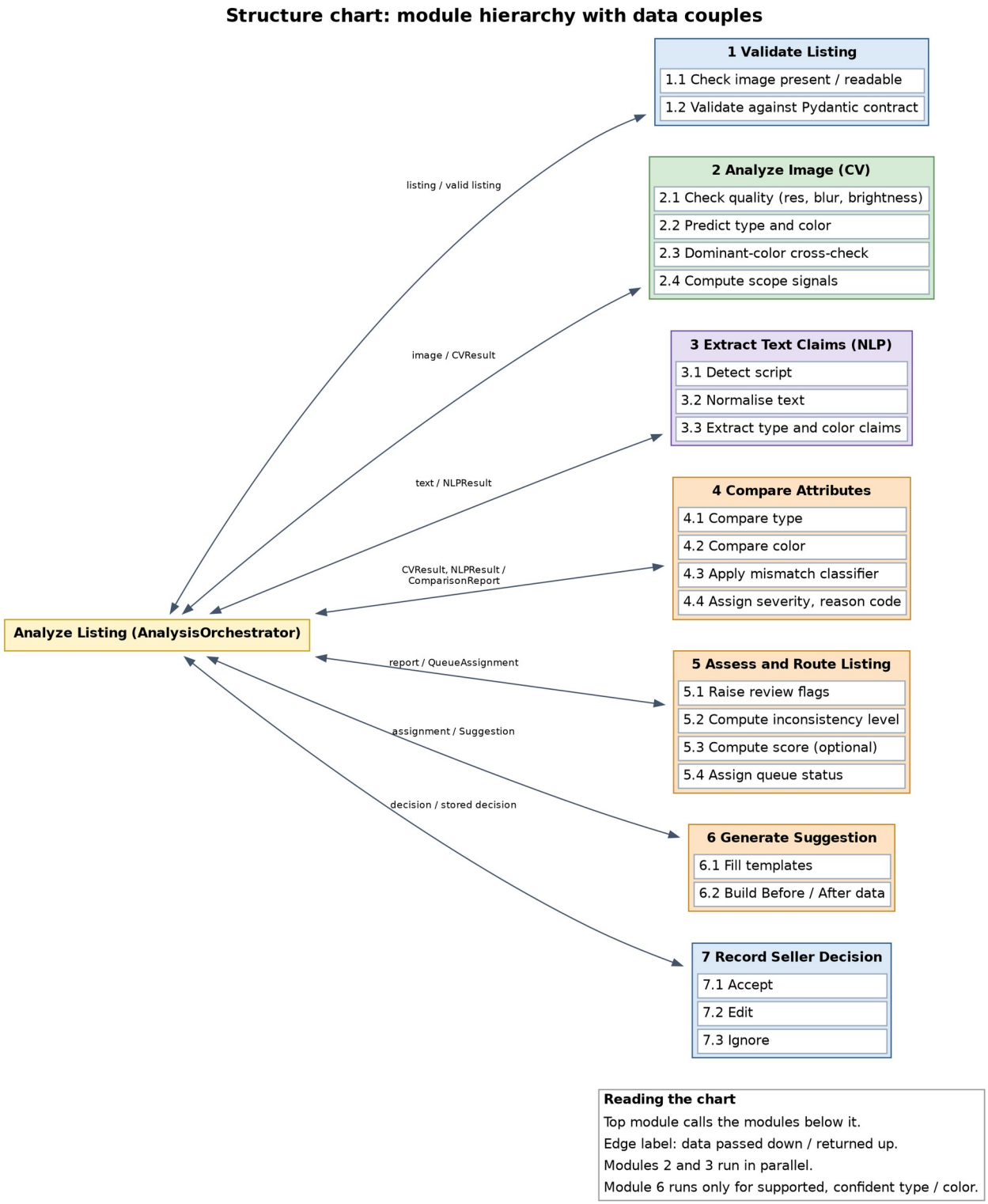


Figure 34. *Structure chart: module hierarchy with data couples*

How to read the diagram

- The top module Analyze Listing (AnalysisOrchestrator) calls seven modules: Validate, Analyze Image, Extract Text Claims, Compare Attributes, Assess and Route, Generate Suggestion and Record Seller Decision.
- Edge labels show data in and data out, for example image to CVResult, or CVResult and NLPResult to ComparisonReport.
- Modules 2 and 3 run in parallel; module 6 runs only for supported, confident type or color; module 5.3 (score) is optional.

Design notes and traceability

Requirements	FR-001 to FR-025, NFR-MNT-01
Objectives	OBJ-01 to OBJ-08

10.10 HIPO Diagram

HIPO (Hierarchy plus Input-Process-Output) documents the same functions as a hierarchy chart plus an IPO table for each top-level function.

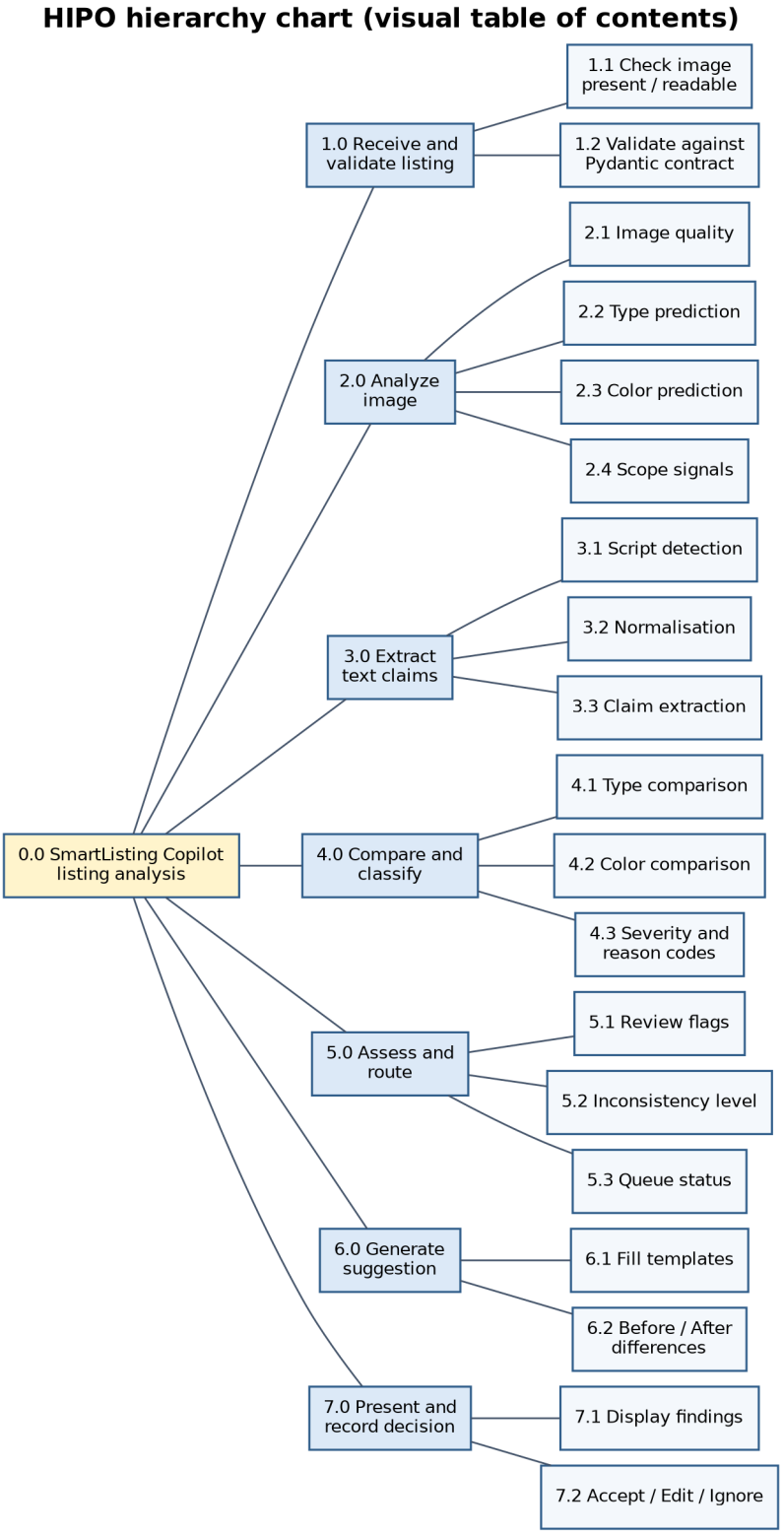


Figure 35. *HIPO hierarchy chart (visual table of contents)*

IPO tables

Module	Input	Process	Output
1.0 Receive and validate listing	Image, title, description, category	Check image present and readable; validate against the Pydantic contract	Valid listing or safe error message
2.0 Analyze image	Valid image	Quality checks; predict type and color; dominant-color cross-check; scope signals	CVResult: labels, confidence, quality flags, scope signals
3.0 Extract text claims	Title, description	Detect script; choose lexicon; normalise; extract type and color claims	NLPResult: claimed type, claimed color, language route
4.0 Compare and classify	CVResult, NLPResult, category	Compare type and color; apply rules and classifier; assign severity; map reason codes	ComparisonReport with findings
5.0 Assess and route	CVResult, ComparisonReport	Raise review flags; compute level; optional score; assign queue status	QueueAssignment, level, optional score
6.0 Generate suggestion	CVResult, NLPResult, assignment	Fill type-and-color templates; build change items	Suggestion or "no definitive suggestion"
7.0 Present and record decision	AnalysisResult, seller choice	Display findings and Before / After; record Accept, Edit or Ignore	Displayed results; SellerDecision (nothing published)

Design notes and traceability

Requirements	FR-001 to FR-025
Objectives	OBJ-01 to OBJ-08
Consistency	The seven top-level functions match DFD processes 1.0 to 8.0 (process 8.0 is merged with 7.0 here) and structure chart modules 1 to 7.

10.11 CRUD Matrix

The CRUD matrix shows which use case creates (C), reads (R), updates (U) or deletes (D) each data entity. Configuration and model weights are read-only at run time.

Use case	Listing	CV	NLP	Findings	Queue	Score	Sugg.	Decis.	Config	Weight	Result
UC-01 Submit Listing	C R										C
UC-02 Analyze Image	R	C							R	R	
UC-03 Extract Text Claims	R		C						R		
UC-04 Compare Attributes		R	R	C					R		
UC-05 Uncertain / Unsupported		R		R	U				R		
UC-06 Missing Attributes (opt.)		R	R	C					R		
UC-07 Assign Queue Status				R	C				R		U
UC-08 Quality Score (opt.)		R		R		C					U
UC-09 View Findings		R	R	R	R	R					R
UC-10 View Suggested Fix	R			R			R				
UC-11 Generate Suggestion		R	R		R		C		R		
UC-12 Decide (12a, 12b, 12c)							R	C			U
UC-13 Review Queue				R	R						R

Legend: C create, R read, U update. "Result" is the AnalysisResult aggregate in the session store. No use case deletes data (D): the in-memory session simply ends. UC-12 creates a SellerDecision and never writes to any marketplace. Entities map to the data model (sections 8.1 and 10.3).

Requirements	FR-001 to FR-025, NFR-SEC-01, NFR-MNT-04
Objectives	OBJ-01 to OBJ-08
Finding	Only UC-07, UC-08 and UC-12 write to the session result after creation; reviewers (UC-13) have read-only access, consistent with the absence of reviewer accounts.

10.12 Gantt Chart

The Gantt charts, taken from the Project Management document [3], show the master schedule and the implementation window with milestones, internal gates and the critical path.

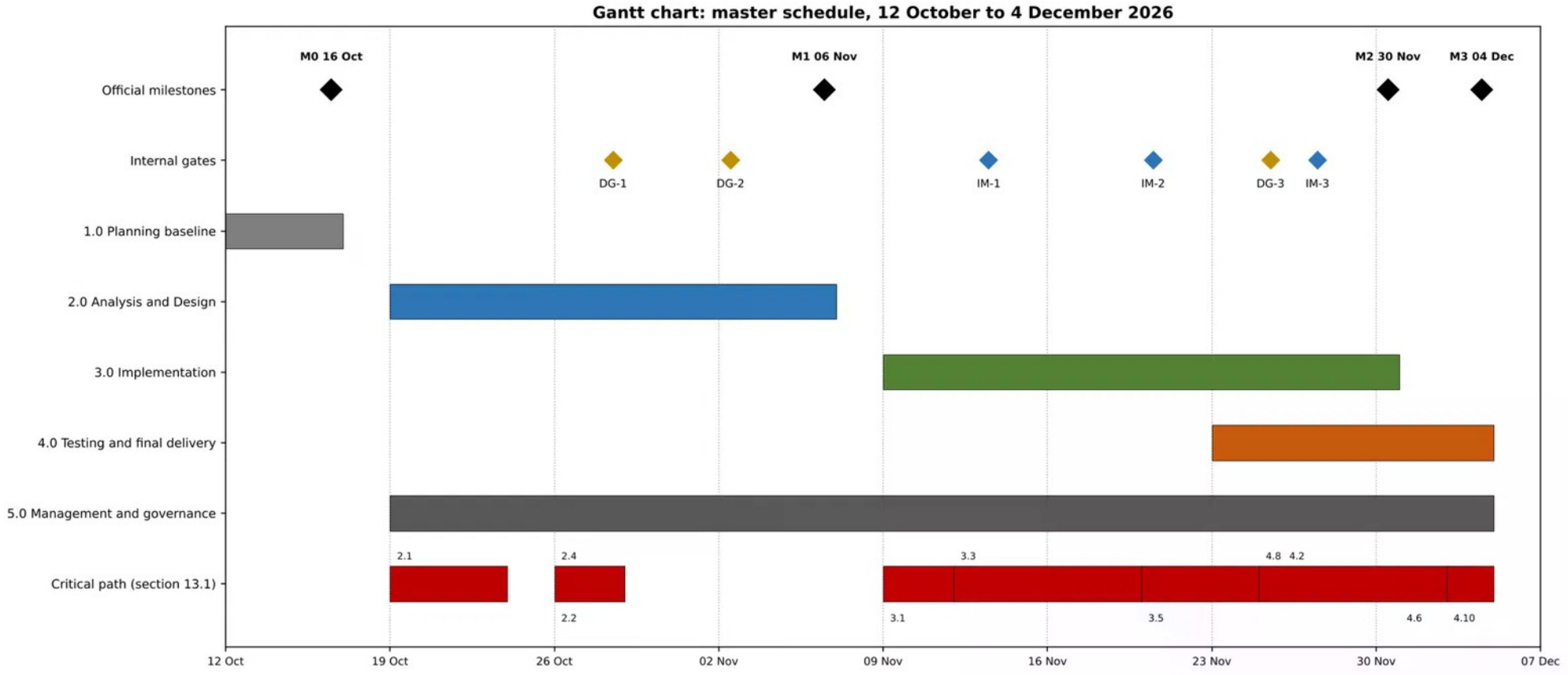


Figure 36. *Gantt chart: master schedule, 12 October to 4 December 2026*

How to read the diagram

- Official milestones: M0 16 Oct, M1 6 Nov (Analysis and Design, this document), M2 30 Nov (Implementation), M3 4 Dec (Testing and final delivery).
- Internal gates: DG-1 to DG-3 (design and data decisions) and IM-1 to IM-3 (integration). Red bars are the critical path.

Gantt chart: Area 3.0 Implementation

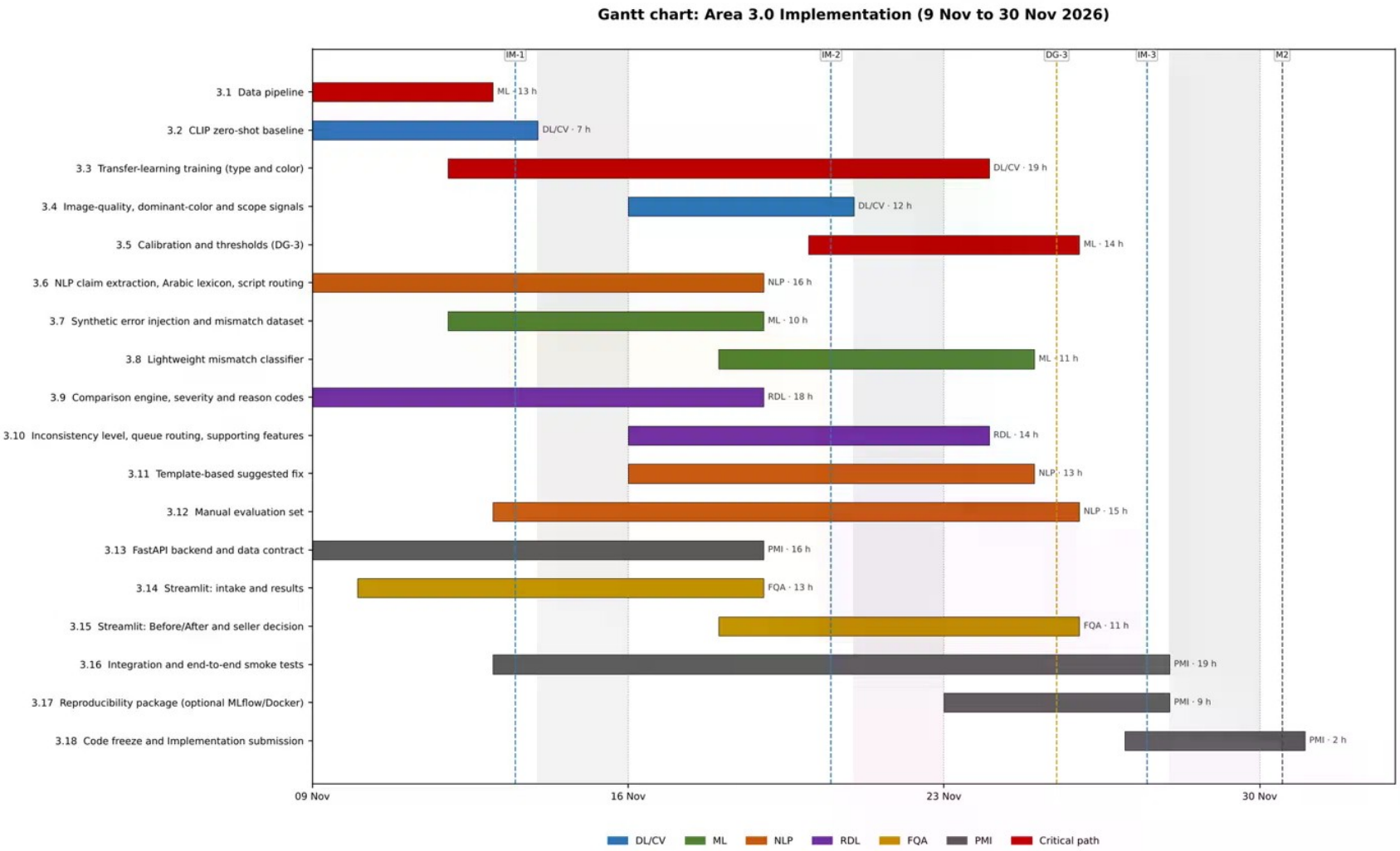


Figure 37. *Gantt chart: Area 3.0 Implementation, 9 to 30 November 2026*

Requirements	CON-11, NFR-PERF-03 (target set after first timing)
Objectives	OBJ-09
Source	Dates, roles and hours follow [3]: 44 work packages, 588 forward hours.

10.13 PERT Chart (PERT Network Diagram)

The PERT network shows the dependency structure of the critical and near-critical work packages as an activity-on-node network.

PERT network (activity-on-node): critical path and near-critical chains, 19 Oct to 4 Dec 2026

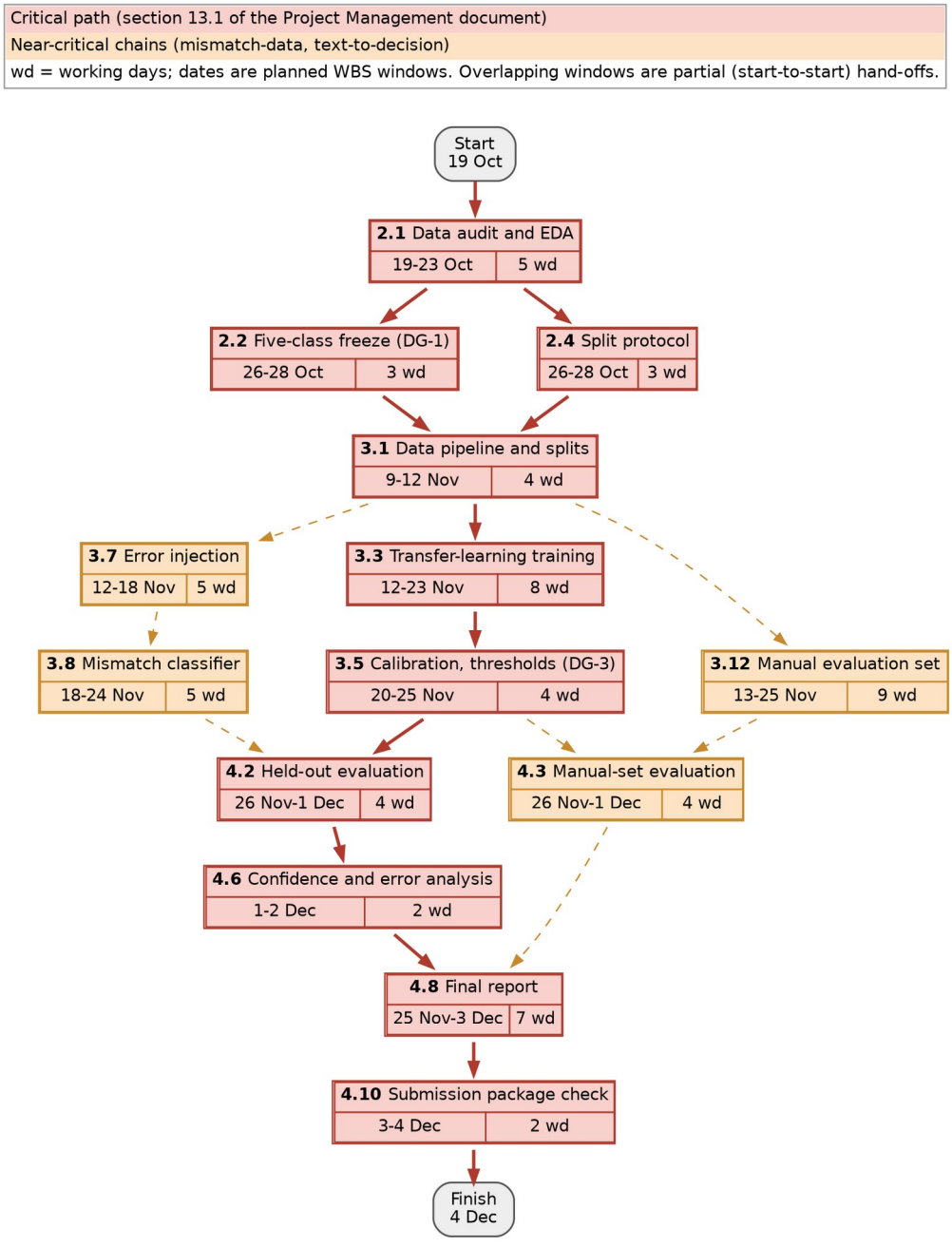


Figure 38. *PERT network: critical path and near-critical chains*

How to read the diagram

- Red nodes and arrows are the critical path of section 13.1 of [3]: 2.1, 2.2 and 2.4, 3.1, 3.3, 3.5, 4.2, 4.6, 4.8, 4.10.
- Orange nodes are near-critical chains: mismatch data (3.7, 3.8) and the manual set (3.12, 4.3).

- Each node shows the WBS number, name, planned window and working days (wd). Overlapping windows are partial hand-offs.

Critical path and slack discipline

Step	WBS	Window	Why it cannot slip
1	2.1 Data audit and EDA	19 to 23 Oct	Class and label decisions depend on it
2	2.2 Five-class freeze (DG-1)	26 to 28 Oct	Splits, vocabulary and models need the class list
3	2.4 Split protocol	26 to 28 Oct	Splitting must precede error injection
4	3.1 Data pipeline	9 to 12 Nov	Training, calibration and manual set need splits
5	3.3 Transfer-learning training	12 to 23 Nov	Longest critical package (8 working days)
6	3.5 Calibration (DG-3)	20 to 25 Nov	Thresholds frozen before held-out results
7	4.2 Held-out evaluation	26 Nov to 1 Dec	Main evaluation evidence
8	4.6 Confidence and error analysis	1 to 2 Dec	Needed for the limitations
9	4.8 Final report	25 Nov to 3 Dec	Ready one working day before submission
10	4.10 Submission check	3 to 4 Dec	Official date 4 Dec

Design notes and traceability

Method	Durations are deterministic working-day counts from the WBS windows. A three-point PERT estimate (optimistic, most likely, pessimistic) is not drawn because the planning documents give a single estimate per package; it can be added if the team supplies ranges.
Objectives	OBJ-09
Source	[3] sections 13.1 and 13.2

11. Consistency, Traceability and Open Decisions

11.1 Diagram-to-requirement traceability

Diagram	Main FRs	Main UCs	Objectives
Class, object, ERD	FR-004 to FR-025	UC-02 to UC-12	OBJ-02 to OBJ-08
Component, deployment, composite, package	FR-001 to FR-003, NFR-MNT, NFR-SEC-06/07	UC-01	OBJ-01
Profile	NFR-REL-03, NFR-MNT-04/06	All	OBJ-01
Use case	FR-001 to FR-025	UC-01 to UC-13	OBJ-01 to OBJ-08
Activity, sequence, communication, interaction overview	FR-001 to FR-025	UC-01 to UC-13	OBJ-01, OBJ-06, OBJ-08
State machine	FR-014, FR-015, FR-019, FR-020, FR-023 to FR-025	UC-05, UC-07, UC-12	OBJ-06, OBJ-08
Timing	NFR-PERF-01 to 03	UC-01	OBJ-01, OBJ-09
Context, DFD, document flowchart	FR-001 to FR-025	UC-01 to UC-13	OBJ-01 to OBJ-08
Flowcharts, decision tree, decision tables	FR-009 to FR-021	UC-04 to UC-07, UC-11	OBJ-04 to OBJ-07
Structure chart, HIPO, CRUD	FR-001 to FR-025	UC-01 to UC-13	OBJ-01 to OBJ-08
Gantt, PERT	CON-11, NFR-PERF-03	None	OBJ-09

11.2 Cross-diagram consistency checks

Check	Result
The same six reason codes appear in the class, object, ERD, DFD and decision-table models	Consistent
The three queue states and the Medium-only open case appear identically in the state, decision tree, decision table and flowchart B	Consistent
Seven top-level functions match across HIPO, structure chart and DFD (8.0 merged with 7.0)	Consistent
No model contains a publish action, state or interface	Consistent (NFR-SEC-01)
No model contains size, material or price attributes	Consistent (CON-03)
Interaction overview references D11b to D11f, all of which exist	Consistent
Class diagram label "D-07 TBC" has no entry in the open-decision register (which ends at D-05)	Naming mismatch to resolve when the register is next updated
Use case diagram marks UC-12 generalisation (12a to 12c) as in the requirements document	Consistent

11.3 Open decisions that affect the diagrams

ID	Decision	Diagrams affected
Q-01	Queue state for exactly one Major (Medium level)	State, decision tree, DT-3, flowchart B, object, use case UC-07
Q-02	Precedence among conflicting flags	State, decision tree, DT-3, flowchart B
IN-01, IN-02	Empty title or description: reject or treat as absent	Activity, sequence D11b, flowchart A, UC-01
IN-03	Unreadable or corrupt image handling	Activity, sequence D11b, flowchart A
IN-04	Blank or malformed category	Activity, sequence D11b, UC-01
D-01	Final five EDA classes and class groups	Class (rules), DT-1, object (example class), PERT

ID	Decision	Diagrams affected
D-02	Color-family and near-shade tolerance	DT-1, DFD 4.0
D-03	Confidence, blur, brightness thresholds	Activity, sequence D11c, UC-05
D-04	Response-time target	Timing, NFR-PERF-03
D-05	Quality Score weights	UC-08, DFD 5.0, structure chart 5.3

11.4 Assumptions and limitations

- Models describe a proposed proof of concept; no behavior has been implemented or tested.
- The Literature Review file was not available; references are limited to the project documents and the standards listed below.
- Diagrams reused from earlier design work were checked against the requirements text; the Requirements document remains the authority when a conflict appears.
- The session store is in memory; the ERD is a logical model, not a commitment to a database (no database of users exists in the MVP).
- Hardware capability and weight-download access are unverified (A-04, LIM-08).
- Approval and sign-off by team or supervisor are pending, as in the requirements document.

12. Conclusion

The analysis and design set defines who uses SmartListing Copilot, what it must do, how it is built, how data and control flow through it, and how the work is scheduled. All 27 required diagrams are provided, plus supplementary class, sequence, data-flow and flowchart views. The models agree with the requirements baseline, show every open decision as open, and contain no publishing path, so the team can start implementation on 9 November against a single, consistent design.

Appendix A. Diagram Inventory

The 27 requested diagrams are listed with their sections and figure numbers. "Reused" means taken from the earlier design work and re-checked; "Redrawn" means rebuilt for readability with the same content; "New" means created for this document.

No.	Diagram	Group	Section	Figure(s)	Status
1	Class diagram	UML structural	8.1	1, 2, 3	Reused
2	Object diagram	UML structural	8.2	4	New
3	Component diagram	UML structural	8.3	5	Reused
4	Deployment diagram	UML structural	8.4	6	Reused
5	Package diagram	UML structural	8.6	8	New
6	Composite structure diagram	UML structural	8.5	7	Reused
7	Profile diagram	UML structural	8.7	9	New
8	Use case diagram	UML behavioral	9.1	10	Reused
9	Activity diagram	UML behavioral	9.2	11	Reused
10	State machine diagram	UML behavioral	9.3	12, 13	Reused
11	Sequence diagrams (6)	UML behavioral	9.4	14, 15, 16, 17, 18, 19	Main reused; 5 alternates new
12	Communication diagram	UML behavioral	9.5	20	Redrawn
13	Interaction overview diagram	UML behavioral	9.6	21, 22	Reused
14	Timing diagram	UML behavioral	9.7	23	Reused
15	Data flow diagrams (L1, L2 x2)	Analysis	10.2	25, 26, 27	L1 redrawn; L2 new
16	Entity-relationship diagram	Analysis	10.3	28	New
17	Context diagram	Analysis	10.1	24	New
18	Flowchart (A and B)	Analysis	10.4	29, 30	New
19	System flowchart	Analysis	10.5	31	New
20	Document flowchart	Analysis	10.6	32	New
21	Decision tree	Analysis	10.7	33	New
22	Decision tables (DT-1 to DT-4)	Analysis	10.8	tables	New
23	Structure chart	Analysis	10.9	34	New
24	HIPO diagram	Analysis	10.10	35	New (hierarchy and IPO tables)
25	CRUD matrix	Analysis	10.11	table	New
26	Gantt chart (master, implementation)	Analysis	10.12	36, 37	Reused from [3]
27	PERT chart	Analysis	10.13	38	New

Appendix B. Reference Rules

Item	Values and rules
Reason codes	TYPE_MISMATCH, COLOR_MISMATCH, LOW_CONFIDENCE, IMAGE_BLUR, MISSING_ATTRIBUTE, UNSUPPORTED_CATEGORY
Severity	Cross-group type = Critical; same-group type = Major; different-family color = Major; near shade = Minor or ignored
Inconsistency level	None Major or Critical = Low; one Major = Medium; two or more Major, or any Critical = High
Queue states	Ready to publish; Needs review; Critical mismatch (Medium-only and combined flags: open, Q-01 and Q-02)
Cut order	Quality Score, Missing Attribute Detection, MLflow, Docker
Milestones	M0 16 Oct, M1 6 Nov, M2 30 Nov, M3 4 Dec 2026
Gates	DG-1 class freeze, DG-2 data contract, DG-3 thresholds; IM-1 13 Nov, IM-2 20 Nov, IM-3 27 Nov

Appendix C. Glossary

Term	Meaning
CV	Computer vision: image-based prediction of Product Type and Color
NLP	Natural language processing: rule-based extraction of claims from text
CLIP	Zero-shot image-text model used as baseline and fallback
ECE	Expected calibration error, a measure of confidence calibration
EDA	Exploratory data analysis, used to select the five classes
Inconsistency Level	Heuristic Low, Medium or High value from Major and Critical mismatches
Review Queue	Three-state local label that prioritises listings for human review
MVP	Minimum viable prototype approved by the Revised Scope
WBS	Work breakdown structure of the Project Management document
wd	Working days

References

[1] SmartListing Copilot Project Proposal (Revised Scope: Fashion Product Images Small), 12 October 2026.

[2] SmartListing Copilot, Member 1 Project Planning Document (Revised), 16 October 2026.

[3] SmartListing Copilot, Member 2 Project Management Document, version 1.0, 15 October 2026.

[4] SmartListing Copilot, Requirements Documentation, Parts I and II, 16 October 2026.

[5] P. Aggarwal, Fashion Product Images (Small), Kaggle dataset, https://www.kaggle.com/datasets/paramaggarwal/fashion-product-images-small

[6] Object Management Group, Unified Modeling Language (UML) Specification, version 2.5.1, December 2017.

[7] K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, 2016.

[8] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML, 2021.